DP

SELECTED RESEARCH / QUARTERLY REPORT

DIMITRIS PECHLIVANIDIS

Stacks

Q3 2026 Benchmark

July 1 to September 30, 2026

Work for Alea Research

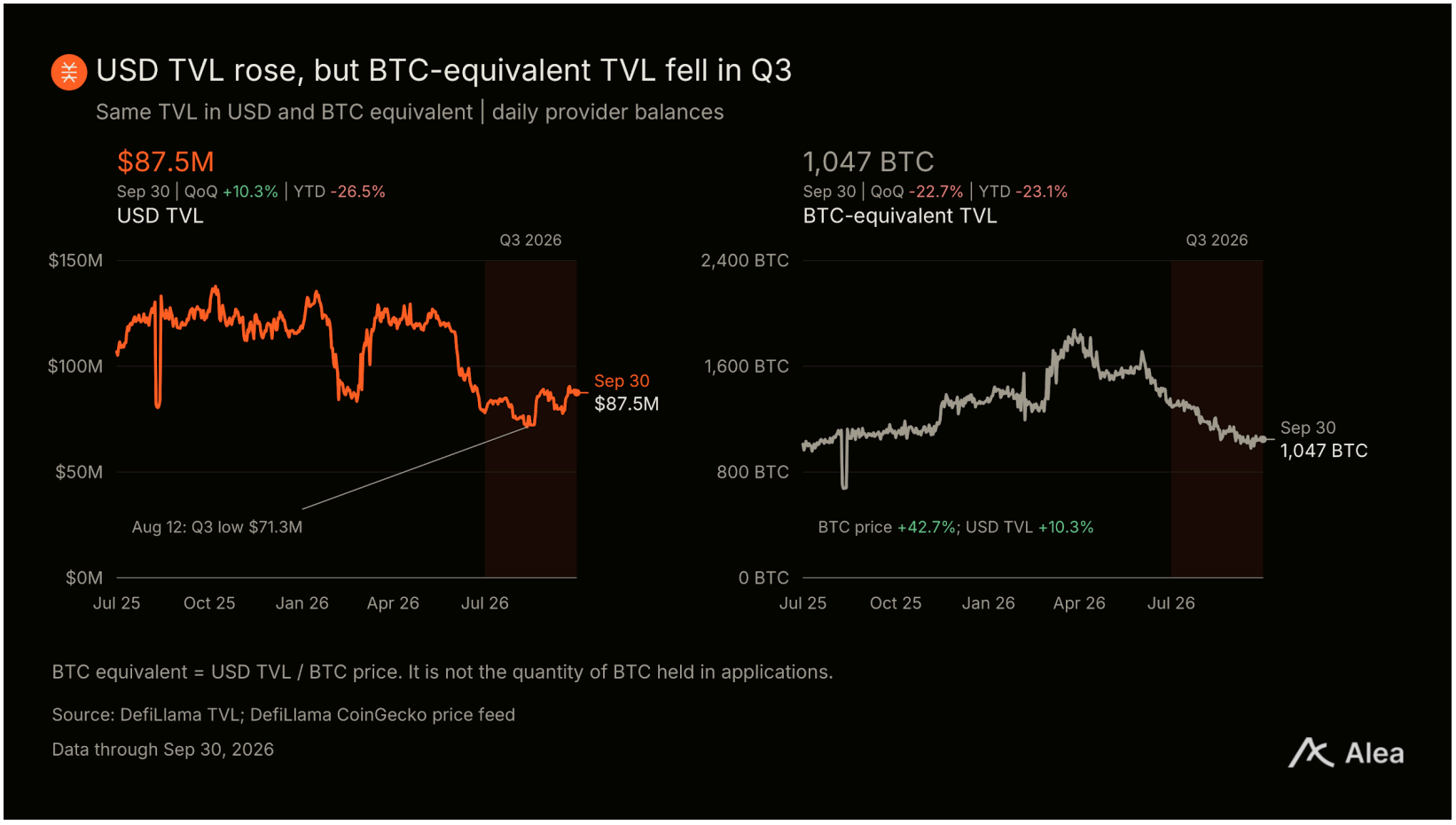
Alea Research was commissioned by Stacks for this report.

The report pages, source links, and disclosure are preserved.

Stacks Q3 2026 Benchmark

July 1 to September 30, 2026

Stacks launched Bitcoin Staking in Q3, adding a funded route for BTC holders and a new reason to lock STX. Genesis ended September with **230.17 BTC** bonded and **3.57M STX** paired. DeFi TVL rose **10.3%** QoQ to **$87.5M**, and network fees increased **105.3%** QoQ to **$94.9K**, bringing YTD fees to **$192.2K**. TVL remained **26.5%** below year-end 2025, while its BTC equivalent fell **22.7%** QoQ. sBTC supply contracted **15.6%** and stablecoin balances fell **11.1%** during Q3. Bitcoin appreciation helped lift dollar balances, while leveraged borrowing increased exposure to existing capital.



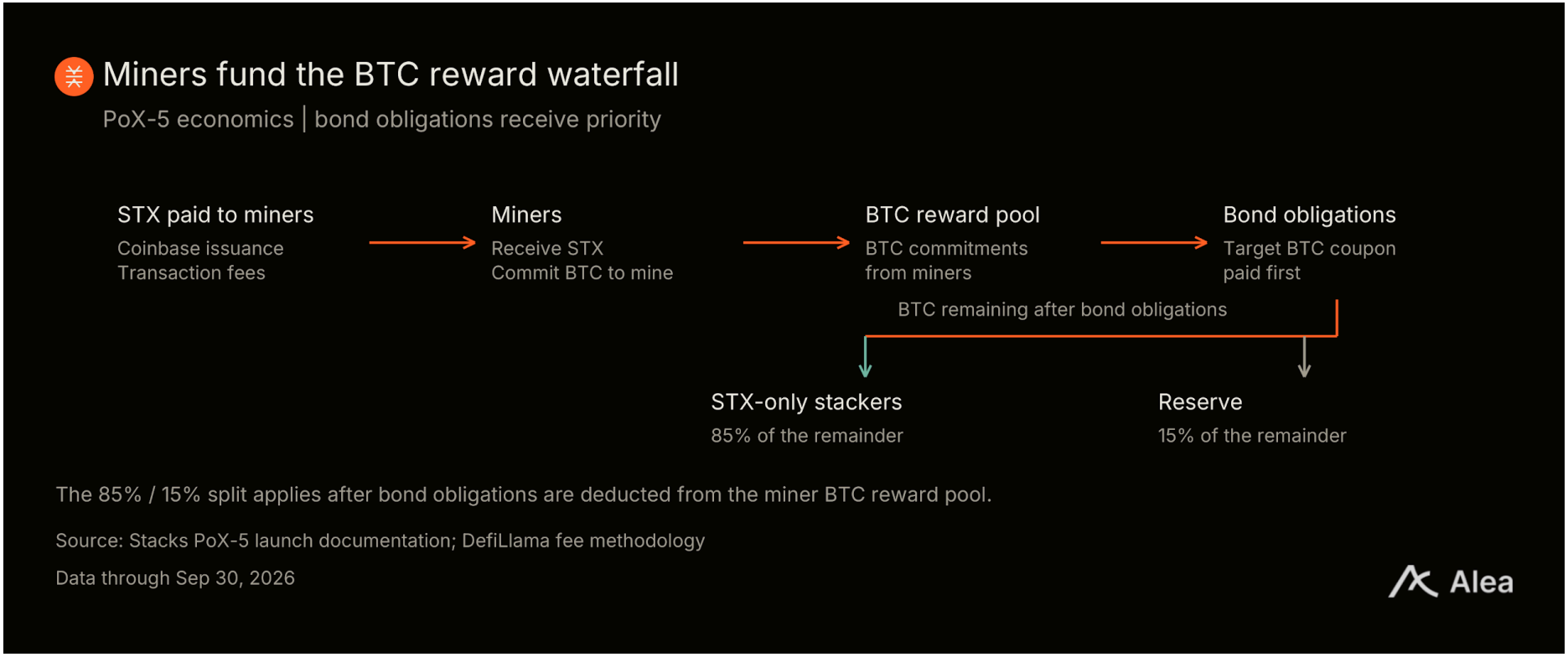
Key Takeaways

- **Bitcoin Staking entered production.** Genesis bonded **230.17 BTC** and paired **3.57M STX**, with 21 calendar days in its live reward period before quarter-end.
- **Dollar TVL recovered while Bitcoin quantities weakened.** DeFi TVL reached **$87.5M**, up **10.3%** QoQ but down **26.5%** YTD. Across four matched applications, Bitcoin appreciation offset a **269 BTC** reduction in direct sBTC cash.
- **Leverage linked Bitcoin Staking to lending.** Zest debt reached **$14.6M**, up from **$6.4M** at June 30. Its Bitcoin Staking vault repeatedly borrowed and staked sBTC, while STX incentives supported selected lending markets.
- **Fees rose as transaction counts fell.** Network fees totaled **$94.9K**, up **105.3%** QoQ, while recorded user transaction types fell **50.7%**.
- **STX repriced faster than operating balances recovered.** STX returned **130.5%** in USD and **61.5%** against BTC during Q3. Its **$694.9M** quarter-end market cap represented approximately **1,098x** the trailing-30-day annualized network fee run rate. Fees accrue to miners.

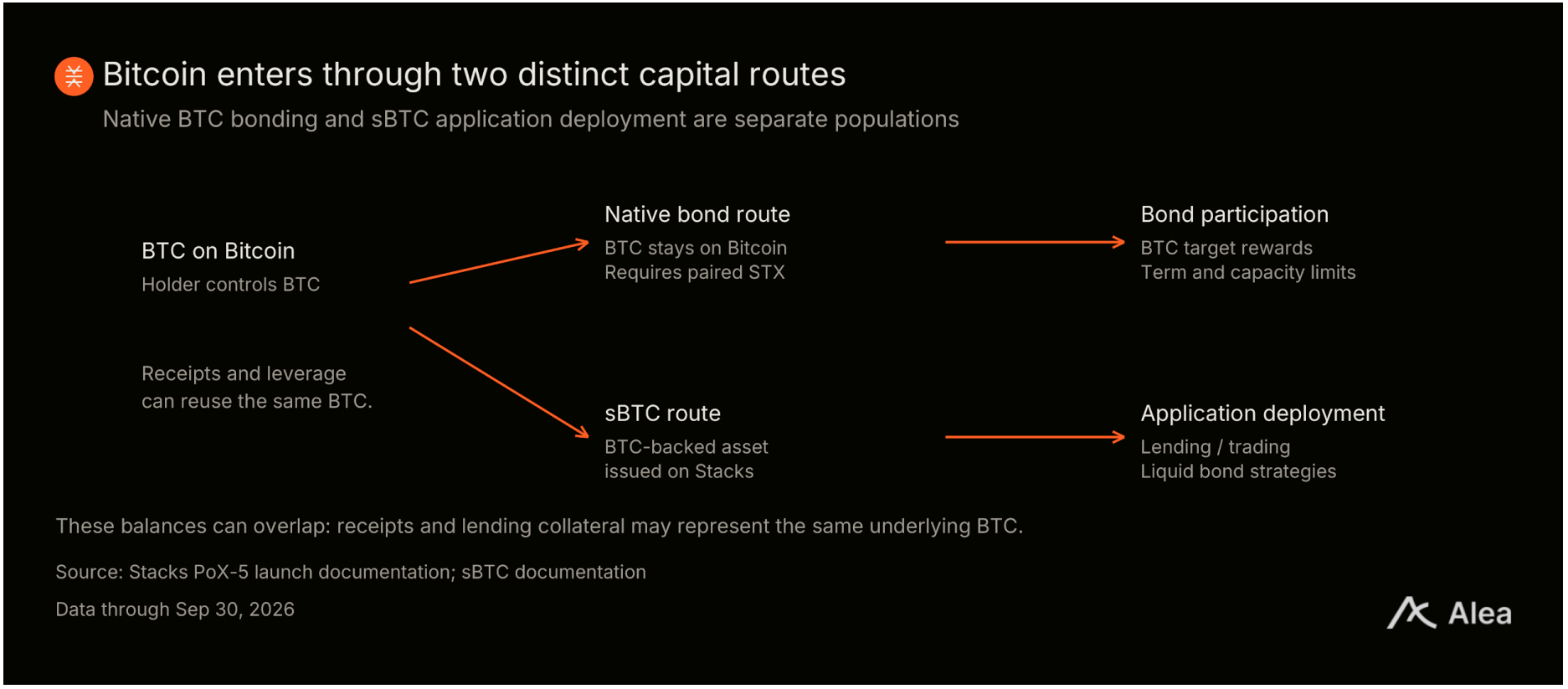
Refresher

Stacks supports smart contracts and financial applications connected to Bitcoin. Users pay transaction fees in STX. Miners commit BTC through Proof of Transfer to compete for STX issuance and transaction fees. Eligible participants receive BTC from that reward system; holding STX alone does not create a claim on every network fee.

PoX-5 extends this mechanism to Bitcoin Bonds. Genesis used a six-month term and targeted approximately **3%** APY, with weekly BTC reward distributions funded by miners. Participants pair BTC with locked STX. Native principal stays in a Bitcoin timelock; the liquid route uses sBTC and depends on its signers. Early BTC exit requires the designated signers' cooperation, while paired STX remains locked for the term. Target yields are not guaranteed.

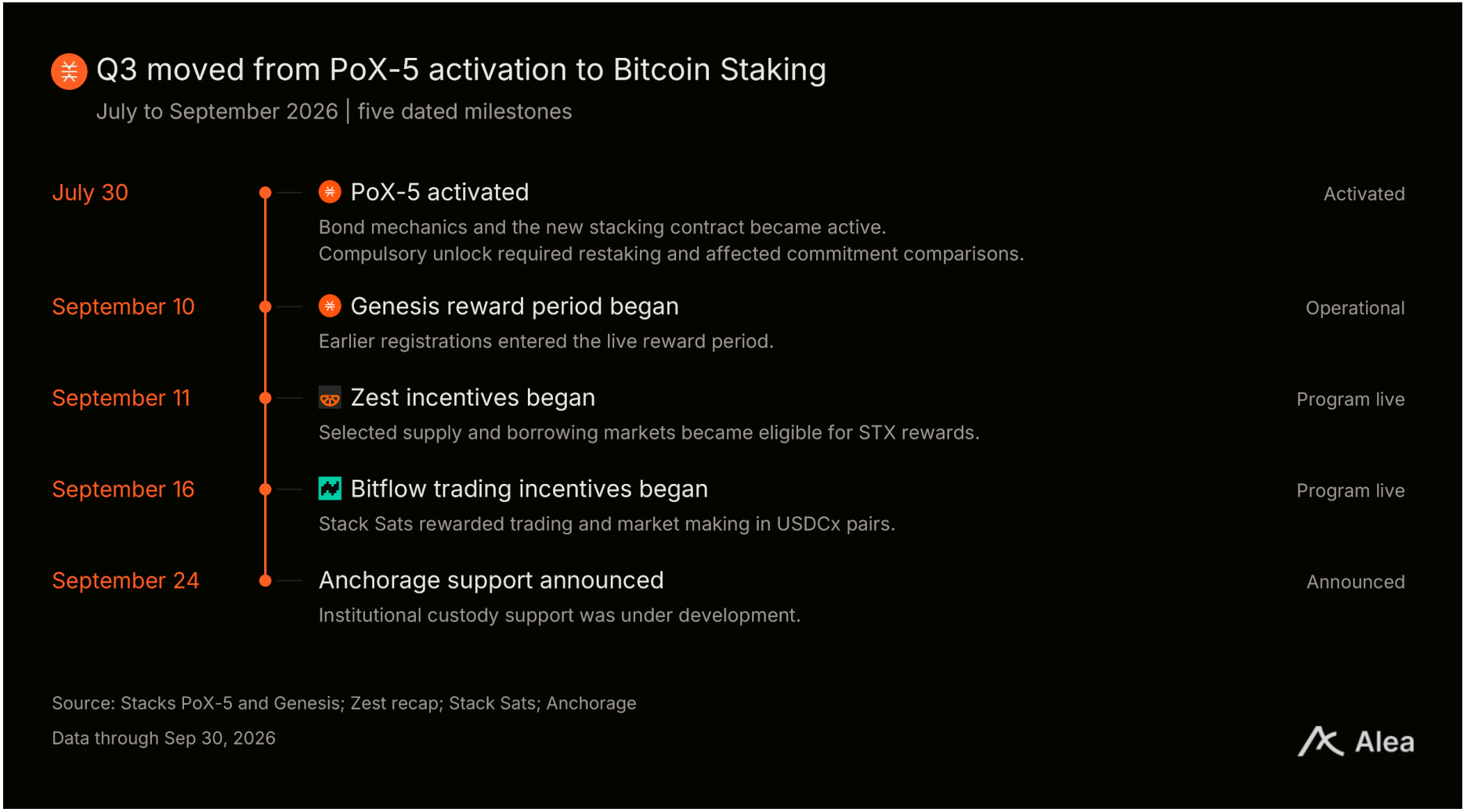


Native bonds keep principal on Bitcoin and sit outside Stacks DeFi TVL. Liquid staking receipts can serve as lending collateral, allowing several positions to reference the same underlying BTC.



Q3 in Context

The quarter crossed two operating regimes. On July 30, **PoX-5** activated. The upgrade unlocked positions in the previous stacking contract. Participants had to restake. Genesis's reward period began on September 10, leaving 21 calendar days inside Q3. Registrations had opened earlier in September.



Anchorage's September announcement described custody support as under development.

Financials

Dollar DeFi TVL rose from **$79.3M** to **$87.5M**, up **10.3%** QoQ but still **26.5%** below year-end 2025. Its BTC equivalent fell **22.7%** QoQ to **1,047 BTC** and **23.1%** YTD.



sBTC supply ended at **2,487 BTC**, down **44.8%** YTD. Stablecoin balances totaled **$12.7M** at a nominal **$1** per unit, down **44.6%** YTD.

Stacked STX reached **448.3M**, down **23%** QoQ and **24.7%** YTD. July's compulsory unlock and contract migration affected this comparison.

Network fees rose to **$94.9K** from **$46.2K** in Q2, bringing YTD fees to **$192.2K**. Annualizing Q3's average daily fees gives **$376.5K**, versus **$185.4K** for Q2; the corresponding YTD annualization is **$257K**.

Spot DEX volume was **$157.1M**, down **2.2%** QoQ, with **$388.2M** traded YTD. Its Q3 daily-average annualization was **$623.2M**, versus **$519M** across YTD.

Supporting financials

Stacks · Q3 2026

Quarter-end balances and quarterly activity · 2025 Q1 to 2026 Q3 · measurement cutoff: September 30, 2026

	2025				2026					
Metric	Q1	Q2	Q3	Q4	Q1	Q2	Q3	QoQ	YTD	YoY
Net sBTC mint / burn (BTC)	2,028	1,985 -43 BTC	-619 -2,604 BTC	109 +728 BTC	-433 -542 BTC	-1,122 -689 BTC	-461 +661 BTC	+661	-2,016	+158
sBTC mint / burn churn (x)	1.13x	0.6x -47%	0.6x +1%	0.09x -84%	0.16x +71%	0.63x +288%	0.36x -42%	-42.3%	1.11x	-40%
DeFi TVL (BTC equivalent)	1,105	1,013 -8%	1,110 +10%	1,361 +23%	1,778 +31%	1,355 -24%	1,047 -23%	-22.7%	-23.1%	-5.7%
Network fees ($, annualized)[1]	≈$309K rounded	≈$195.7K rounded	$140.7K	$94.6K -33%	$207.2K +119%	$185.4K -11%	$376.5K +103%	+103.1%	$257K	+167.5%
Fees (BTC equiv., annualized)[1]	≈3.7 rounded	≈1.8 rounded	1.2	1.1 -12%	3 +181%	3.2 +4%	4.5 +42%	+42.3%	3.1	+265.1%
Spot DEX volume ($, annualized)[1]	≥$243.6M 49/90 days	$809.7M	$784.1M -3%	$444.8M -43%	$286.1M -36%	$644M +125%	$623.2M -3%	-3.2%	$519M	-20.5%

[1] Annualized = quarter flow ÷ calendar days × 365. This differs from the quarter-end 30-day fee run rate used in P/F.
YTD: BTC-equivalent TVL shows change from Dec 31; net flows and churn cover Jan to Sep; annualized flows use the Jan to Sep mean × 365.
BTC equivalents use the quarter-end BTC price; they are neither actual BTC holdings nor fees collected in BTC.
sBTC churn = (gross mint flows + gross burn flows) ÷ full-quarter mean supply (92 days in Q3).
Net-flow QoQ/YoY shows the BTC difference. ≈ = rounded fee totals; ≥ = incomplete coverage, a minimum total. Missing history is not zero.

Source: DefiLlama; Stacks team; CoinGecko prices via DefiLlama; Token Terminal · Sep 30, 2026

Alea

Bitcoin Capital: Bonds and sBTC

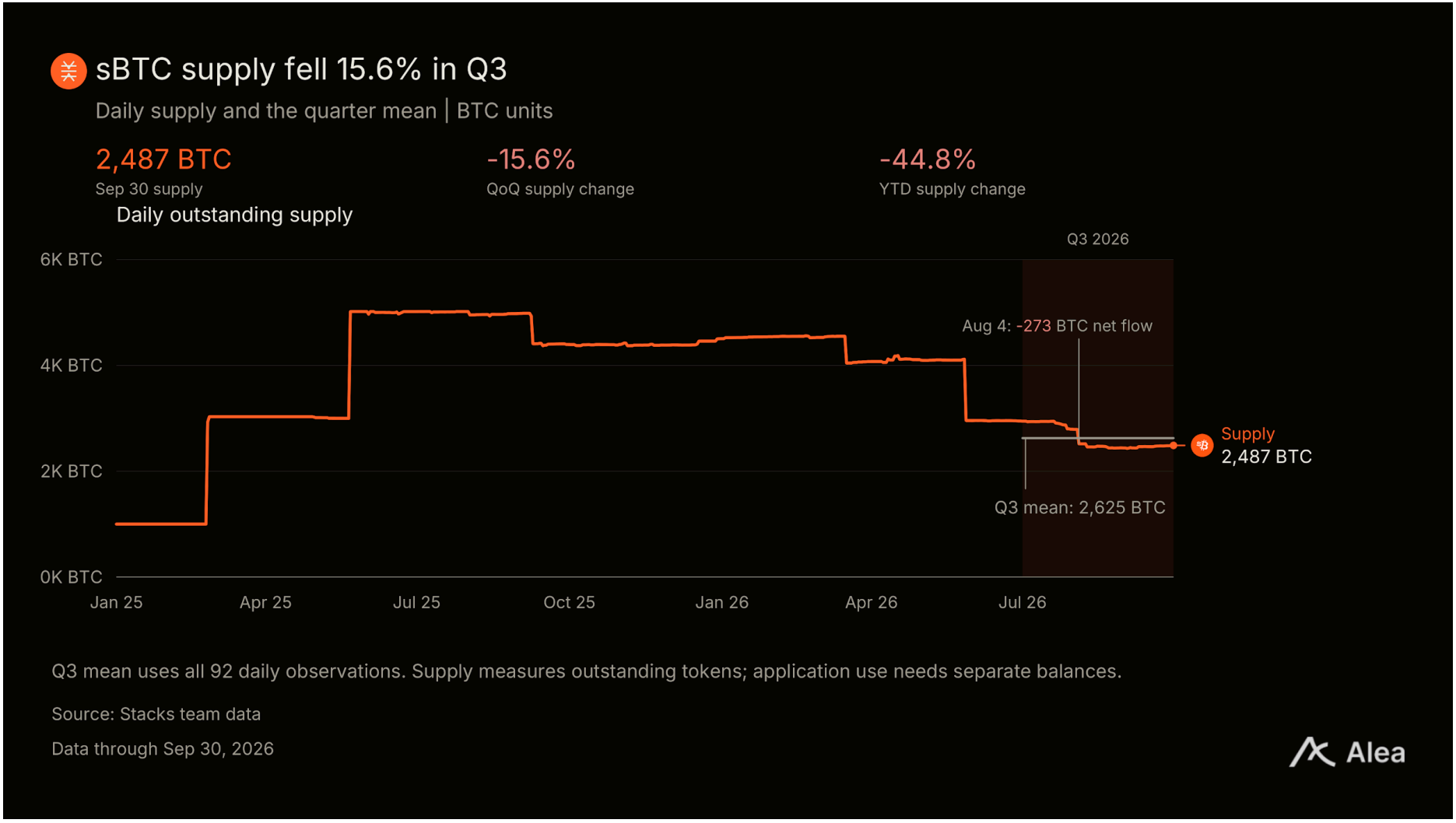
Genesis launches

Genesis ended September with **230.17 BTC** of principal: **70 BTC** in native Bitcoin timelocks and **160.17 BTC** through the liquid sBTC route. Participants paired **3.57M STX** with those positions. The **15 registrations** included pooled positions. Institutional participants included UTXO Management, 21Shares, and HashKey on the native route, plus Sypher Capital through StackingDAO. Genesis was deliberately sized as a small initial rollout.

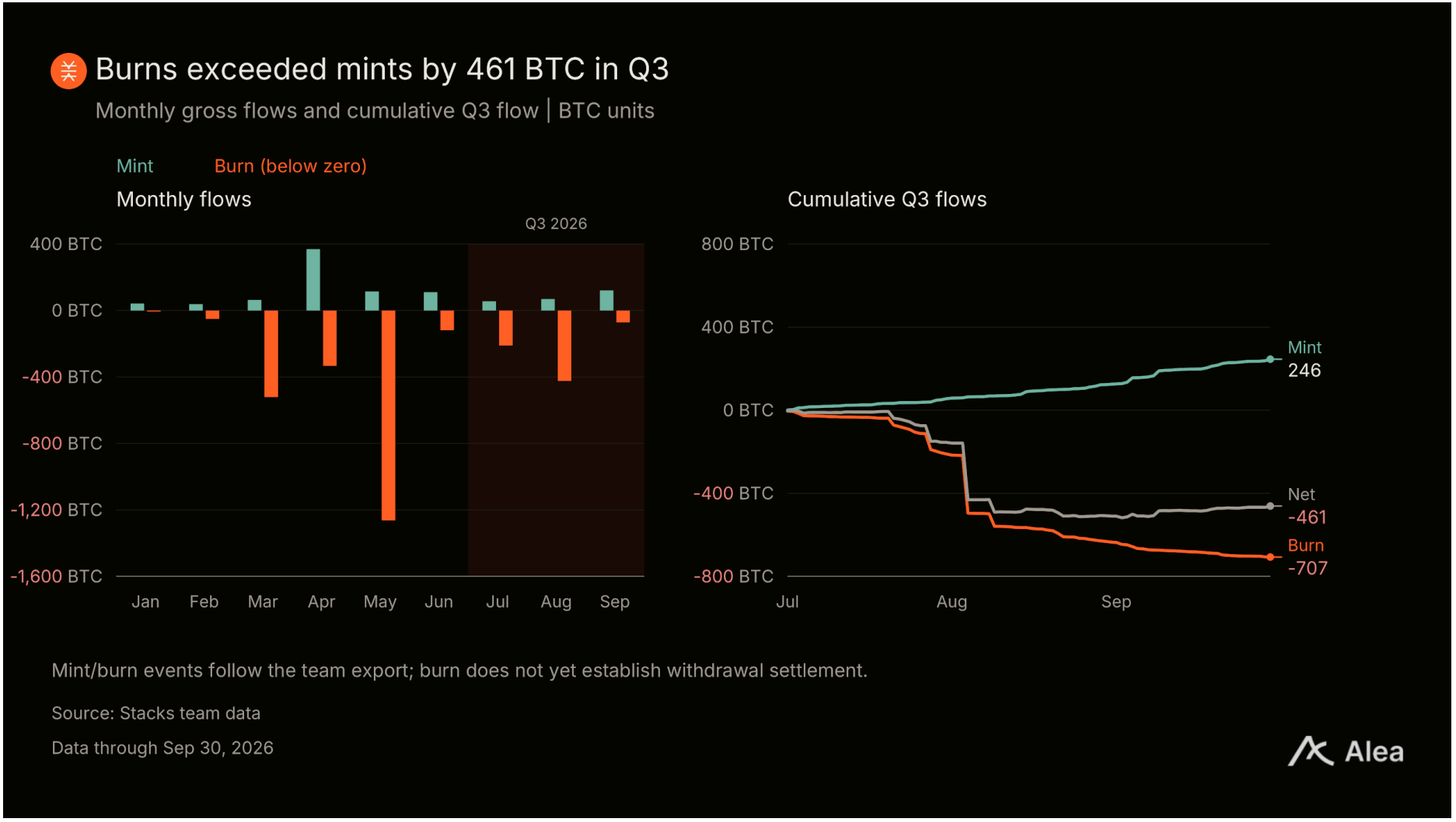
The native route holds BTC on Bitcoin; the liquid route commits sBTC on Stacks. Both require paired STX, while the liquid route connects directly to Stacks applications.

sBTC outflows persist

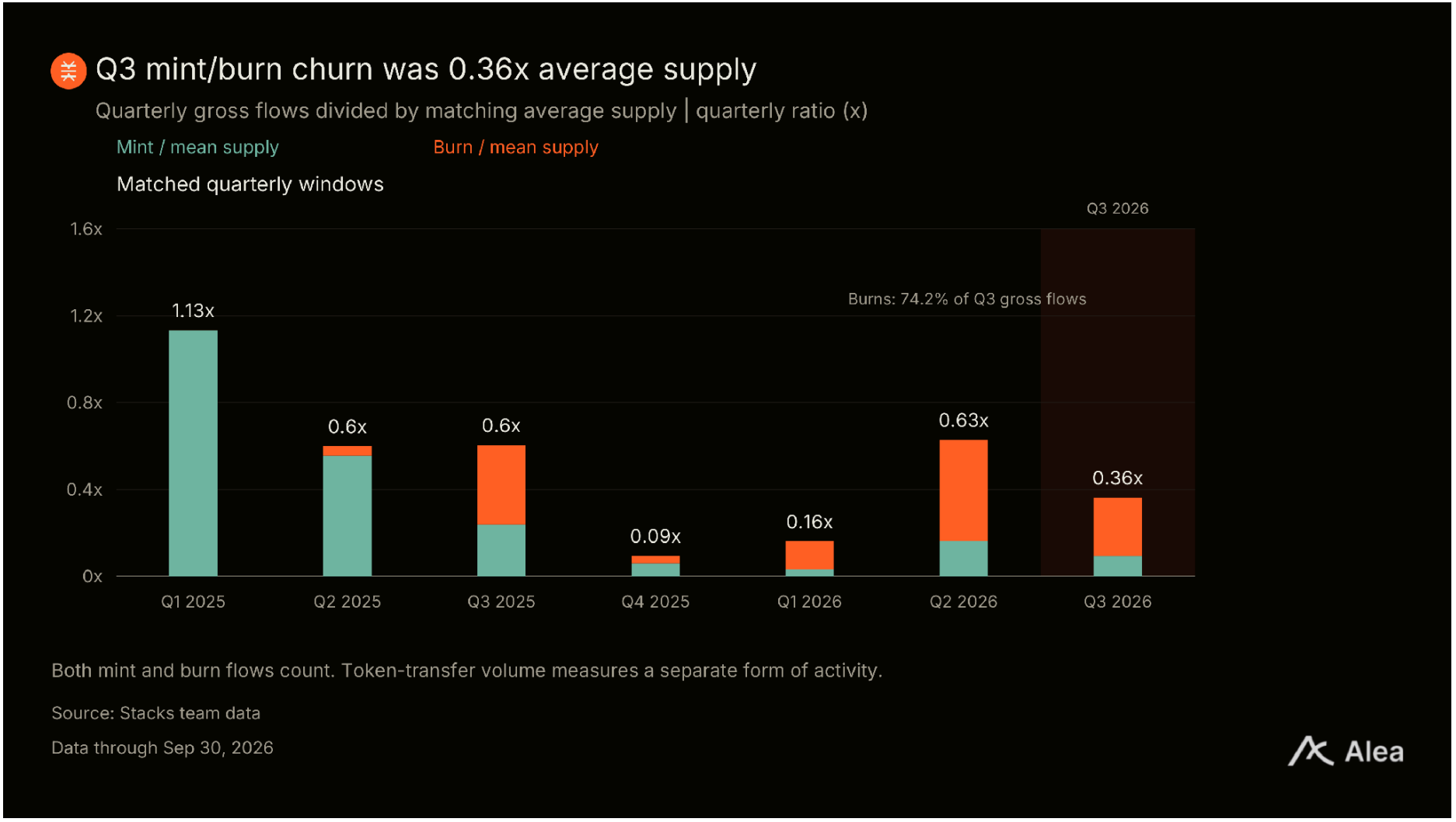
Supply ended **15.6%** below June 30 and **43.4%** below September 2025.



Q3 mints totaled **245.63 BTC** and burns totaled **706.83 BTC**, a net contraction of **461.21 BTC**. That loss was smaller than Q2's **1,121.81 BTC** contraction, but it extended YTD net outflows to **2,016 BTC**.

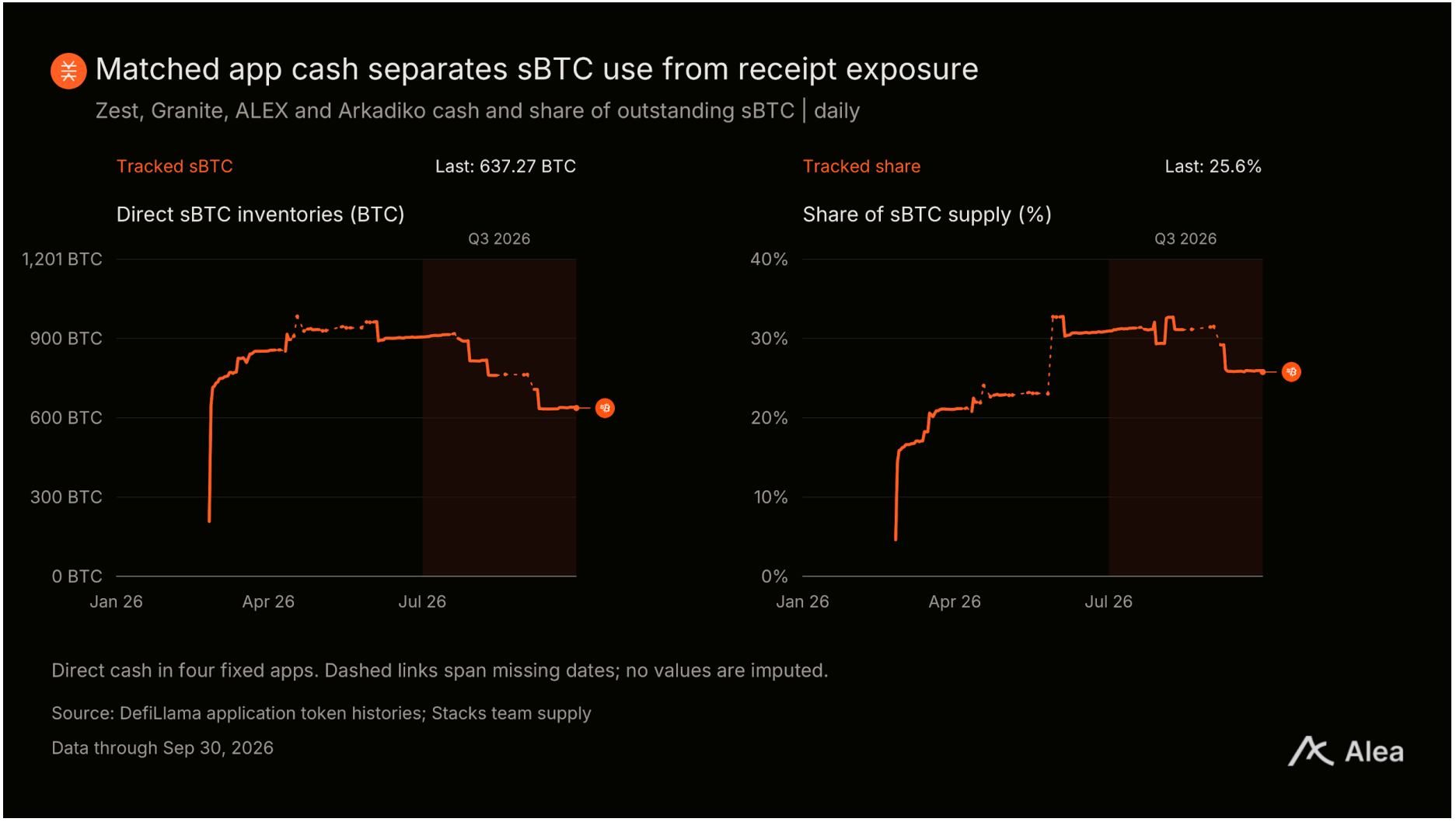


Q3 mean supply was **2,625 BTC** across all 92 days. Mint-and-burn churn fell from **0.63x** in Q2 to **0.36x** in Q3, measuring issuance and burn activity relative to average supply.

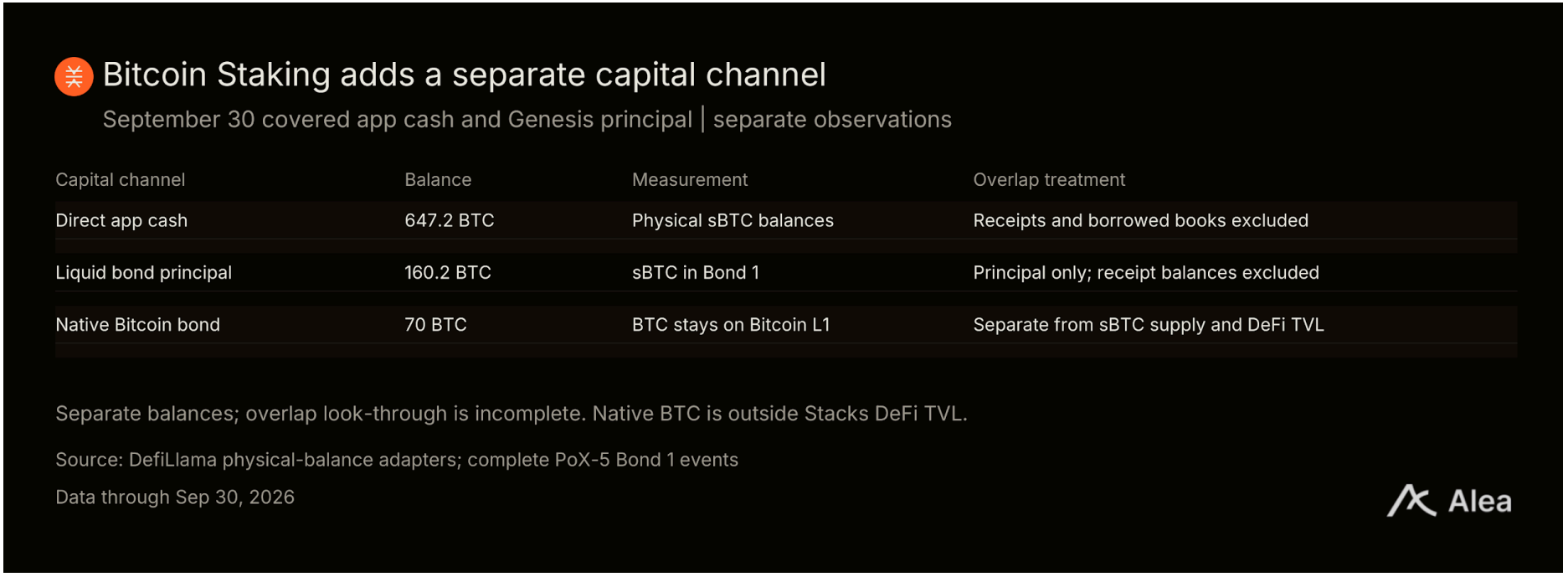


Concentrated deployment

Four applications with matched historical observations held **637.27 BTC** in direct sBTC cash at September 30, equivalent to **25.6%** of outstanding supply. Adding separately reported Bitflow cash brings the covered quarter-end cash inventory to **647.18 BTC**.



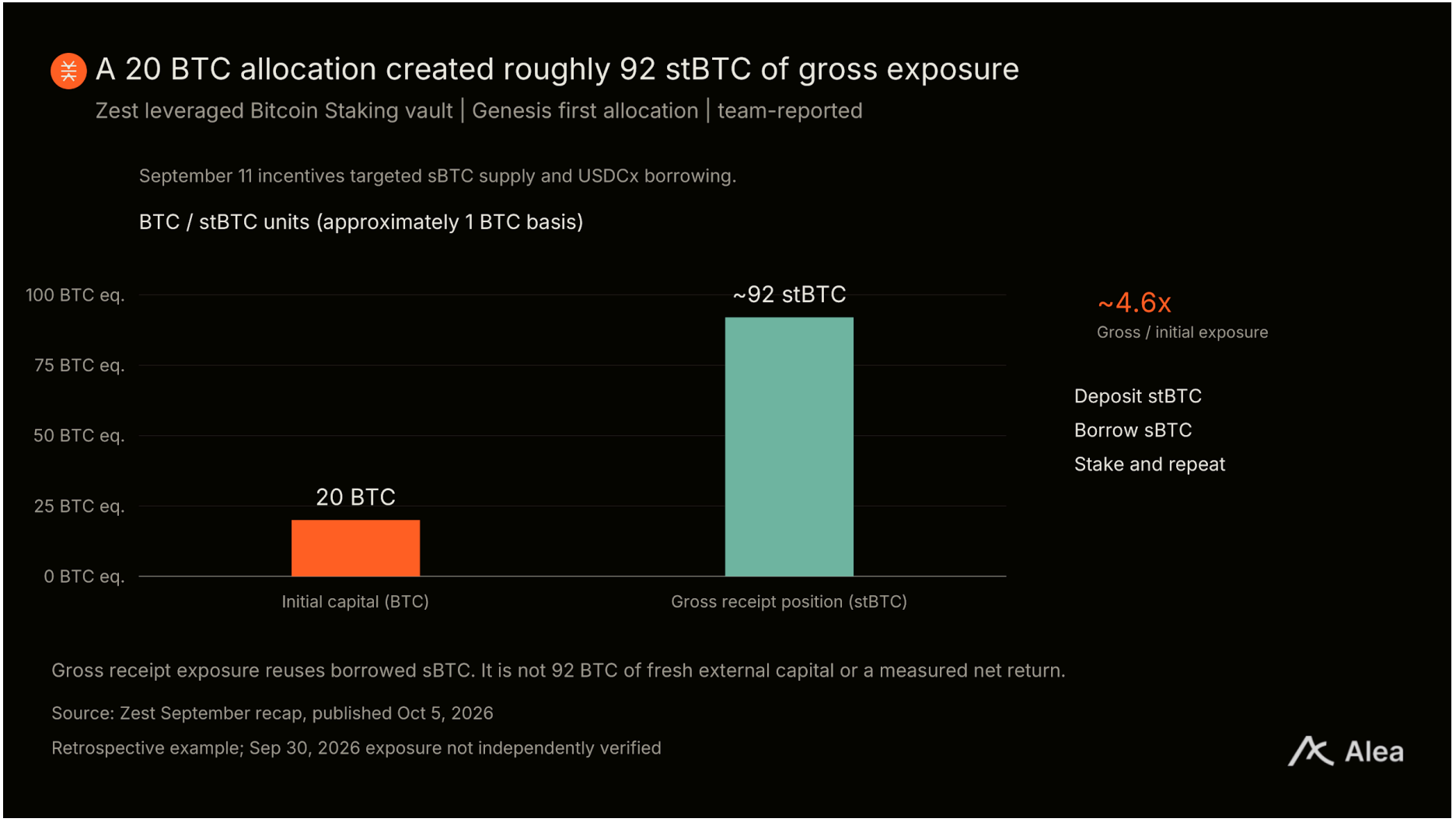
Genesis held **160.17 BTC** of liquid bond principal, separate from the application cash inventory. Receipt assets and collateral can reference BTC already counted elsewhere.



Zest accounted for approximately **598 BTC** of the matched cash inventory, concentrating most covered sBTC cash in one lending venue.



Zest reported that its Bitcoin Staking vault and new incentives drove September borrowing growth. Its October 5 recap described **20 BTC** of initial capital generating roughly **92 stBTC** of gross exposure, or **4.6x**, through repeated borrowing and staking. Borrowing costs and BTC reward rates determine the spread on this strategy.

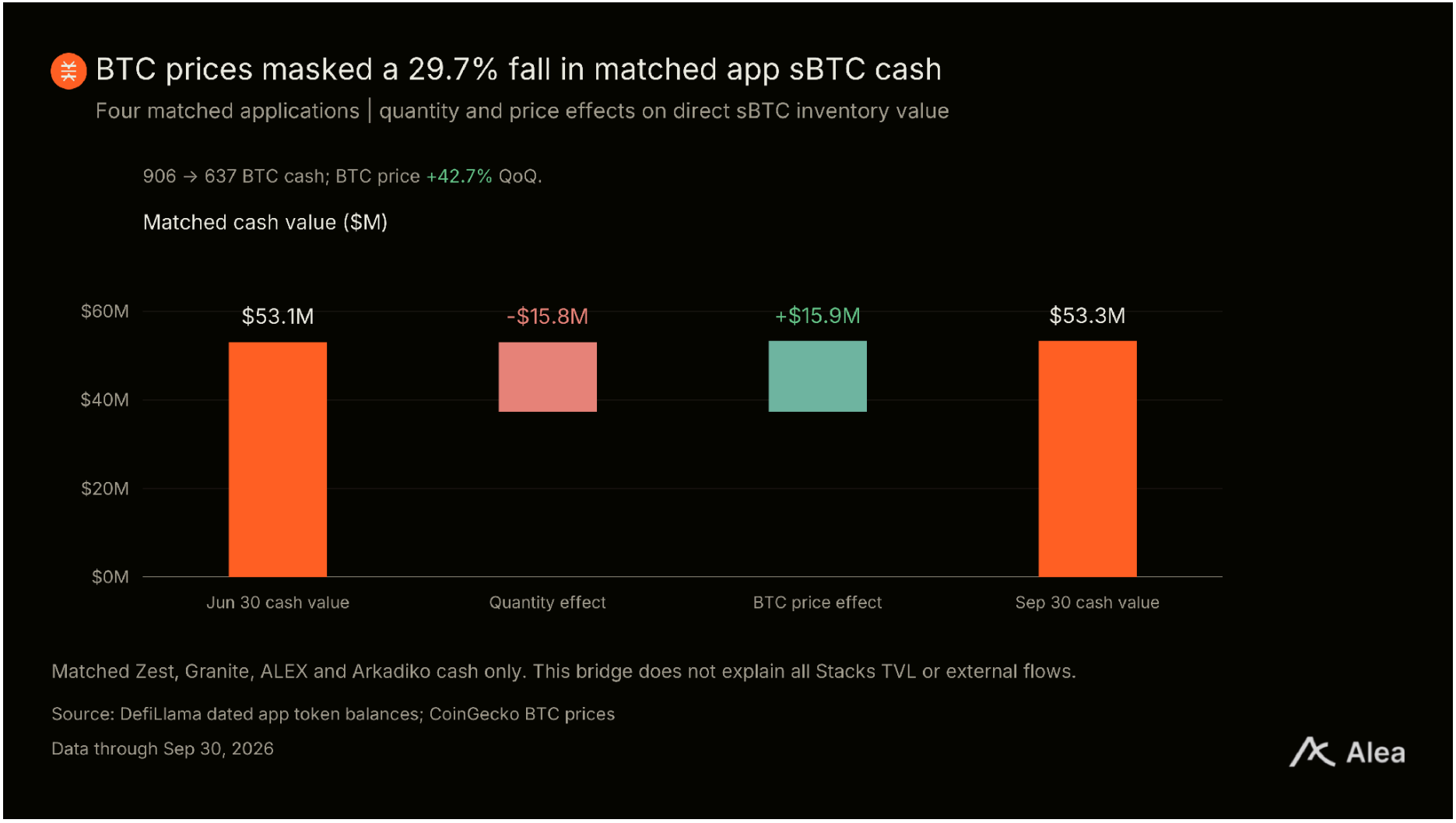


DeFi TVL and Liquidity

Price gains mask lower holdings

Bitcoin rose **42.7%** in Q3, outpacing the **10.3%** rise in dollar TVL. TVL's BTC equivalent consequently fell **22.7%**.

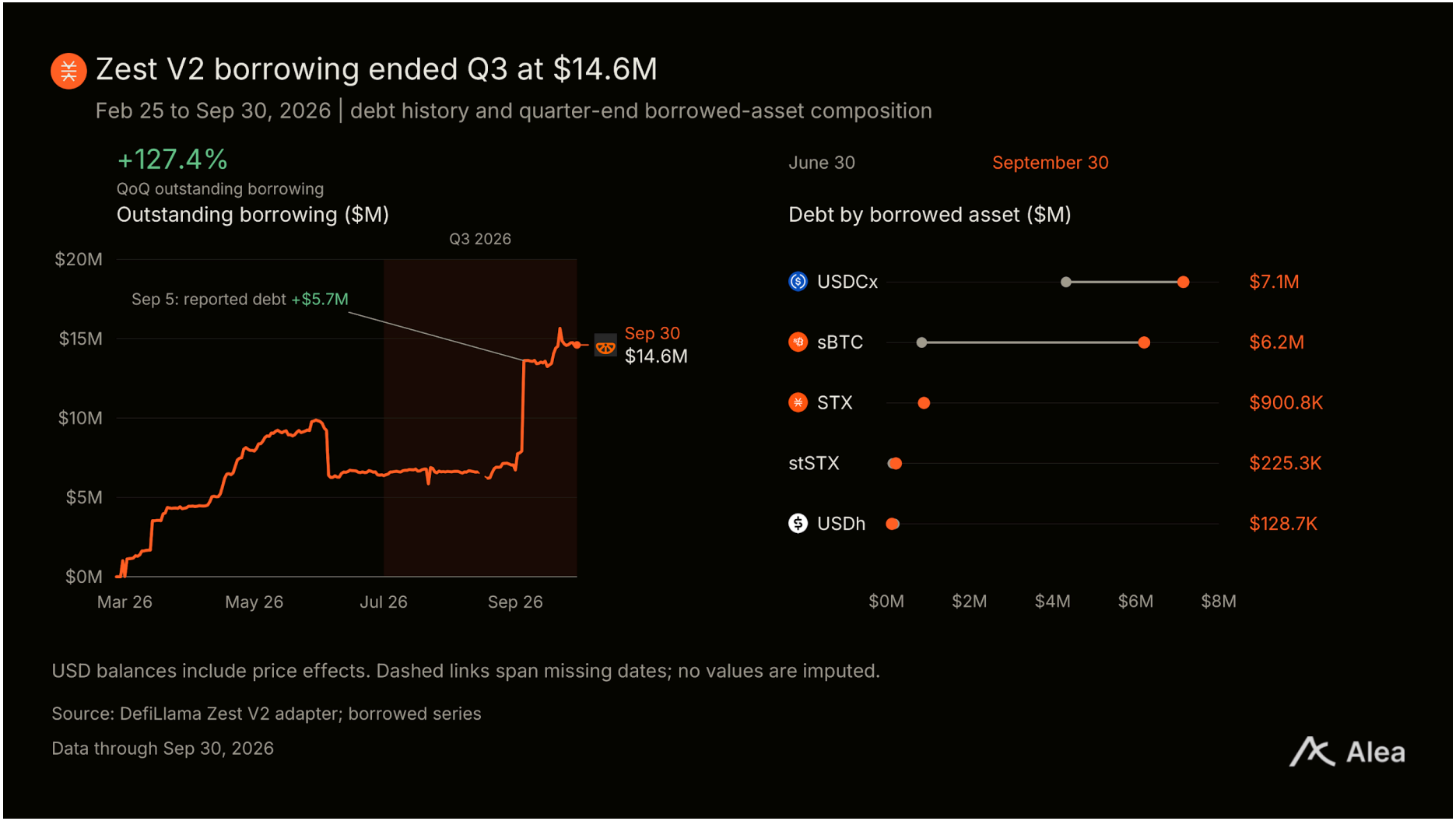
Across the four matched applications, direct sBTC cash fell from **906.2 BTC** to **637.27 BTC**, down **29.7%**. At the opening BTC price, the quantity change removed **$15.75M**. Bitcoin appreciation added **$15.94M** to the remaining balance, leaving dollar cash almost unchanged at **$53.26M**. Cash holdings can fall through borrowing or withdrawals.



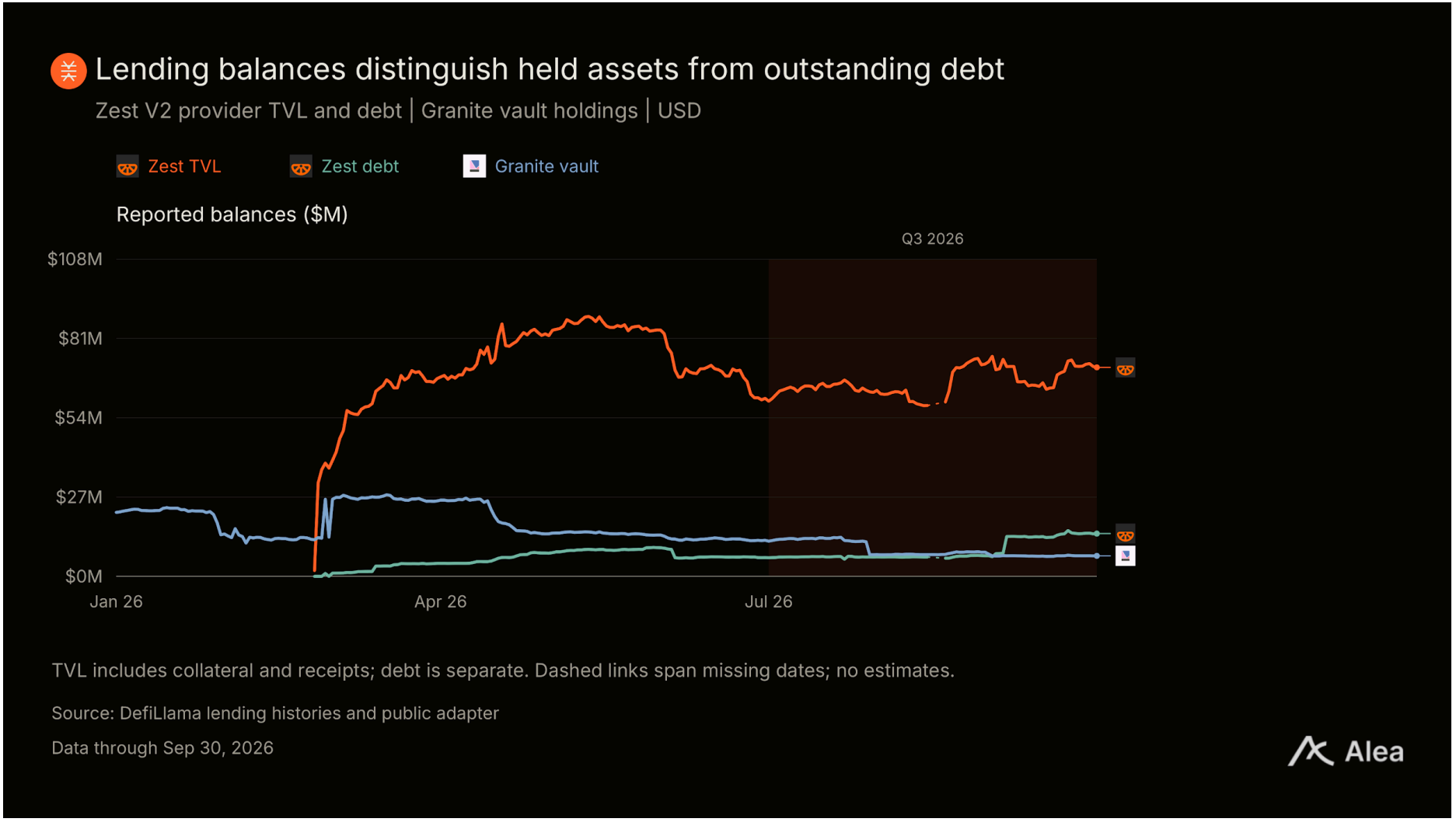
The Endowment deployed approximately **$6.6M** in DeFi liquidity and market-making initiatives, with a significant portion expected to return to the treasury before year-end. Liquidity was split across centralized and decentralized venues to support STX access for Bitcoin Staking.

Borrowing expands

Zest's outstanding debt reached **$14.6M** at September 30, up **127.4%** from **$6.4M** at June 30. Borrowing grew ahead of Genesis's reward period: a **$5.7M** increase on September 5 accounted for **69.7%** of the quarter's net debt growth.



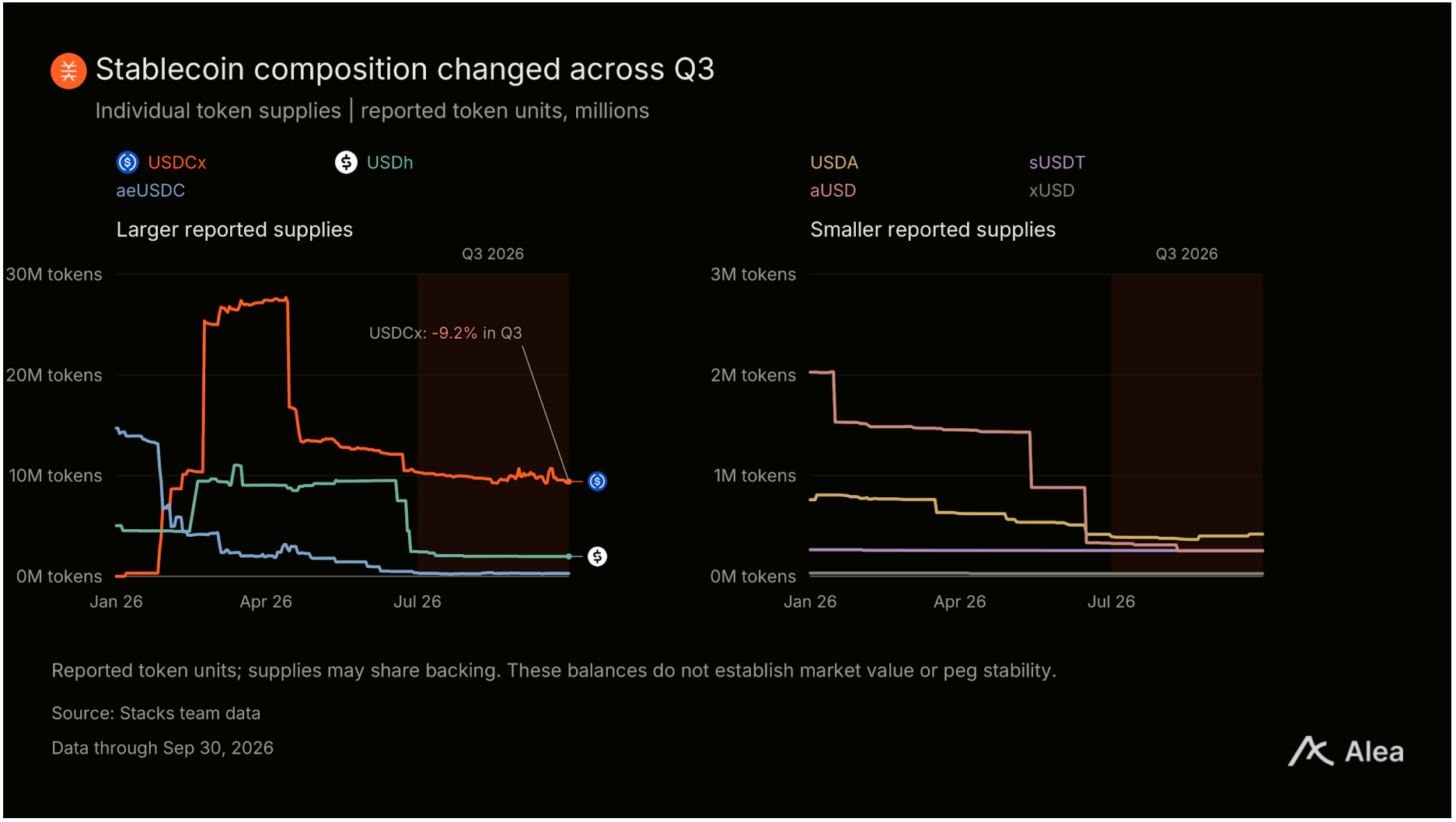
Zest held **$71.3M** in TVL alongside **$14.6M** of debt. TVL includes collateral and receipt assets, so debt divided by TVL is not lending utilization. Granite's **$7M** measures vault holdings.



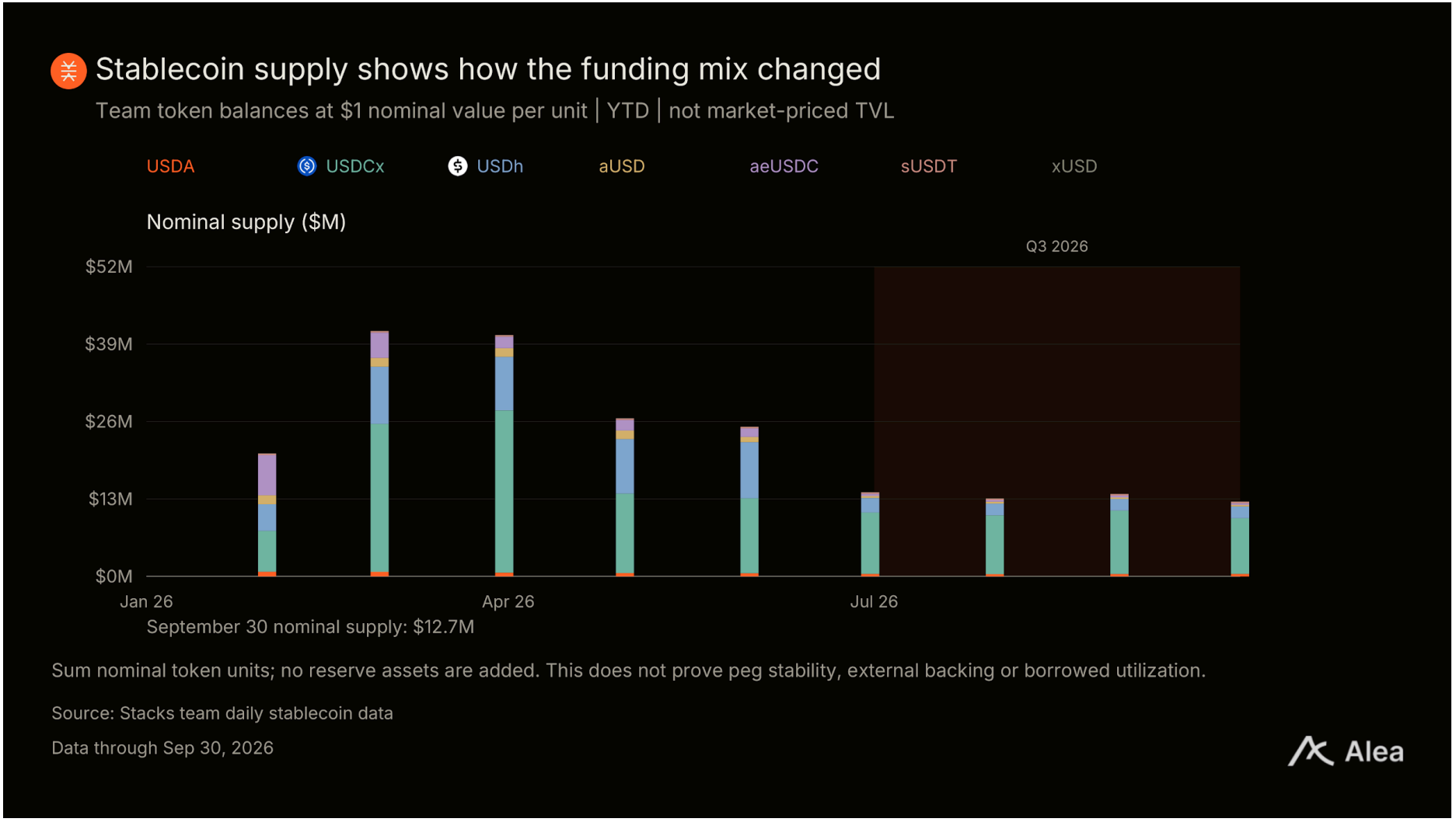
Liquidators repay undercollateralized debt in exchange for discounted collateral. The borrower bears the realized collateral loss.

Stablecoins shrink; USDCx dominates

Stablecoin balances totaled **$12.65M** at a nominal **$1** per unit, down **11.1%** QoQ and **44.6%** YTD.



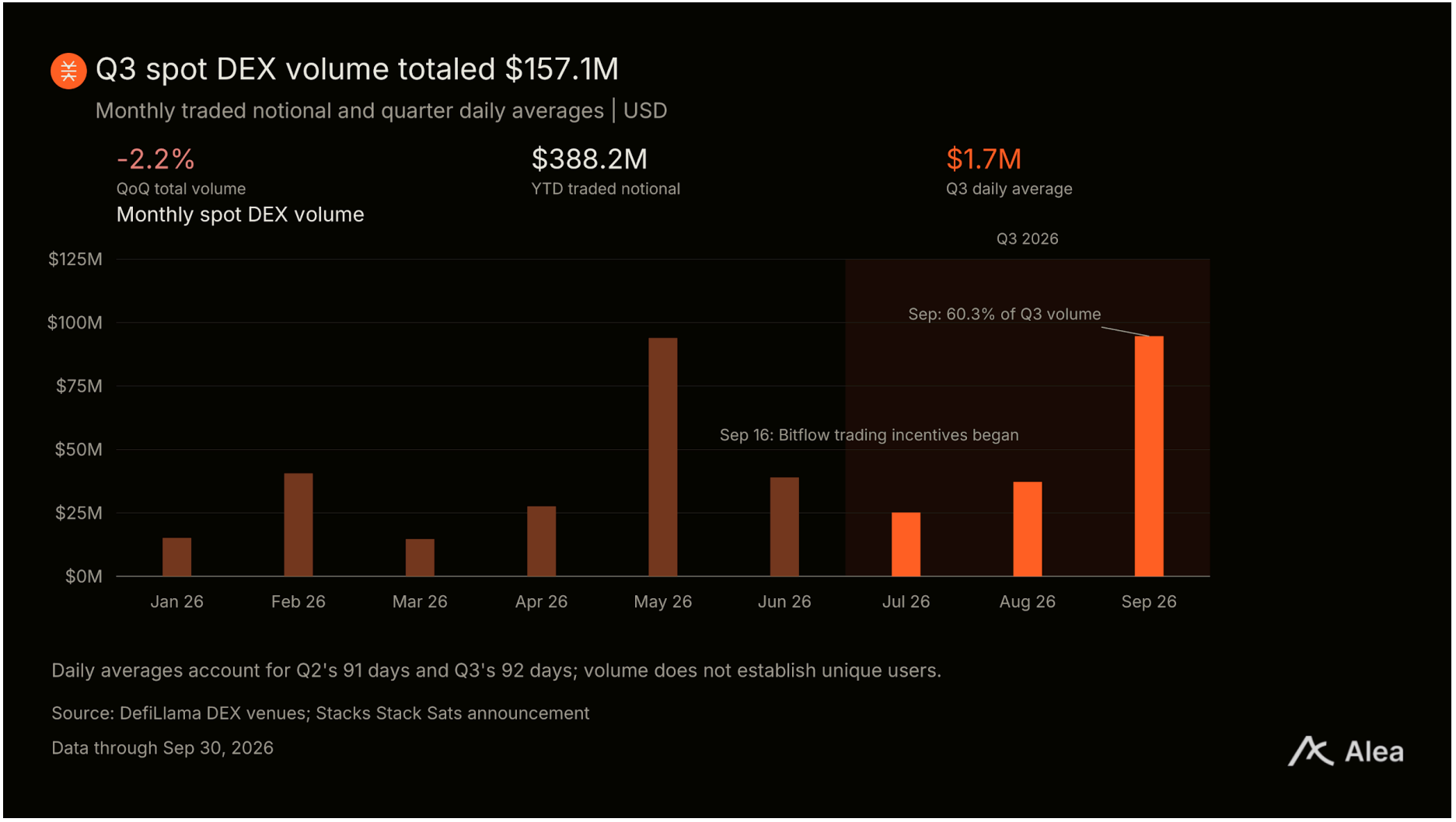
USDCx represented **$9.42M**, or **74.4%** of quarter-end stablecoin balances. Liquidity shifted toward USDCx while the overall pool shrank.



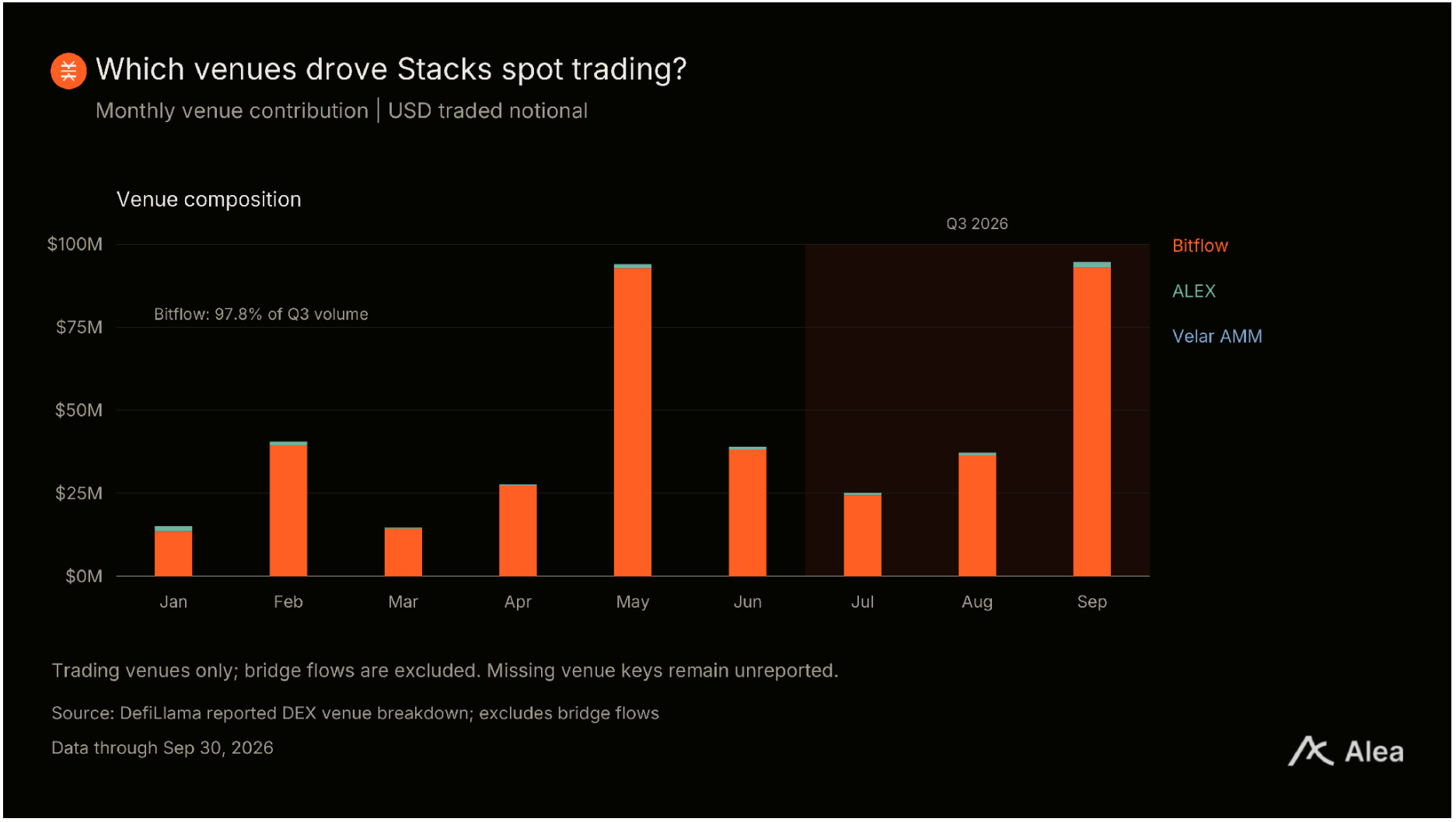
September leads DEX activity

Quarterly spot DEX volume was **$157.1M**, down **2.2%** QoQ and **20.5%** YoY. September generated **$94.6M**, or **60.3%** of Q3 activity.

The quarter's daily average was **$1.71M**, versus **$1.76M** in Q2 and **$1.42M** across YTD.

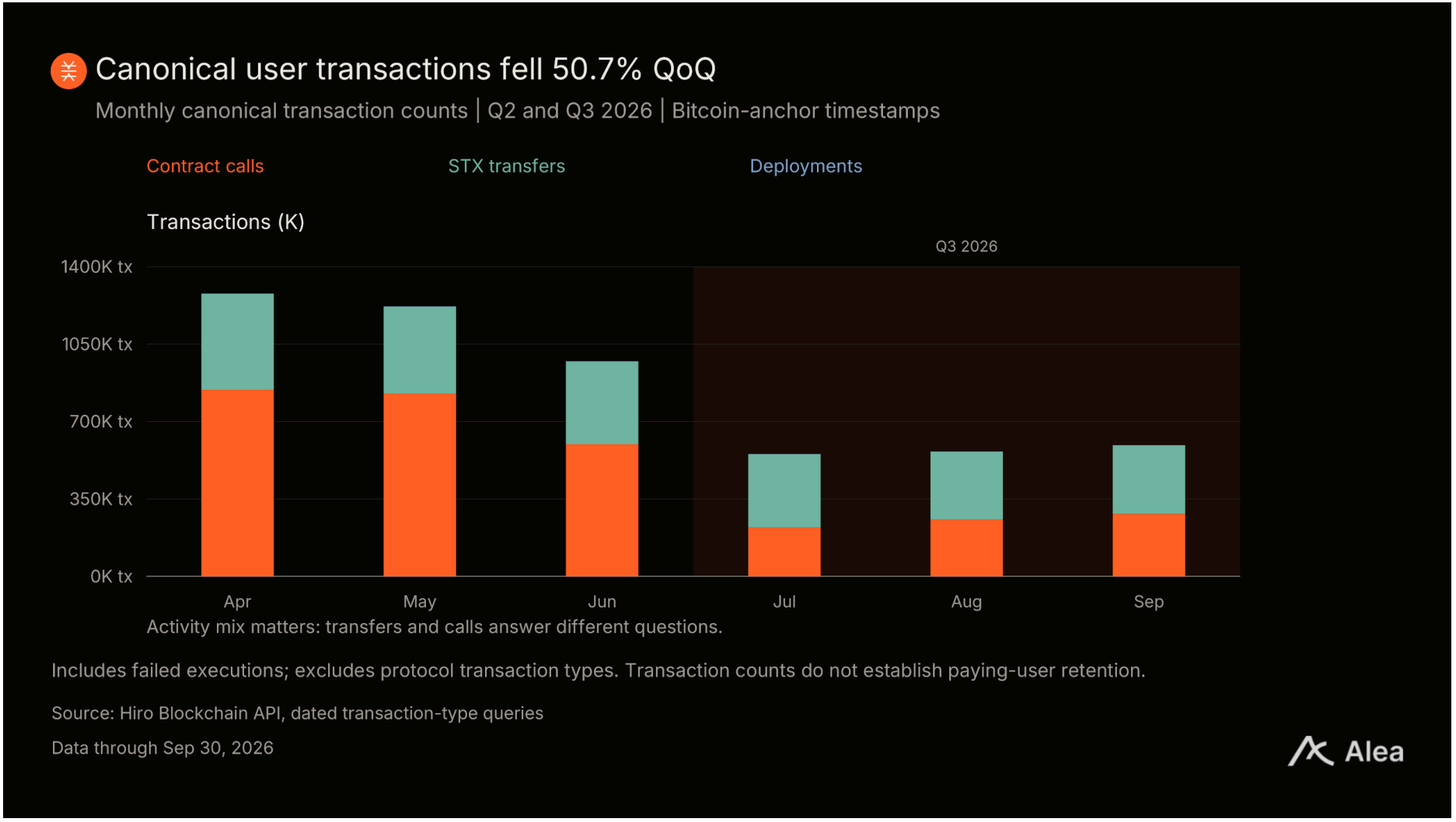


Bitflow generated **$153.7M**, approximately **97.8%** of quarterly volume. Its Stack Sats trading and market-making incentives began September 16 on the sBTC/USDCx and STX/USDCx pairs. The September rebound coincided with that new reward window and remained concentrated in one venue.



Network Activity

Recorded contract calls and STX transfers, together with smart-contract deployments, totaled **1.71M** in Q3, down from **3.47M** in Q2. September's **593.6K** was above August's **564.7K**, but this September improvement did not reverse the **50.7%** QoQ decline.

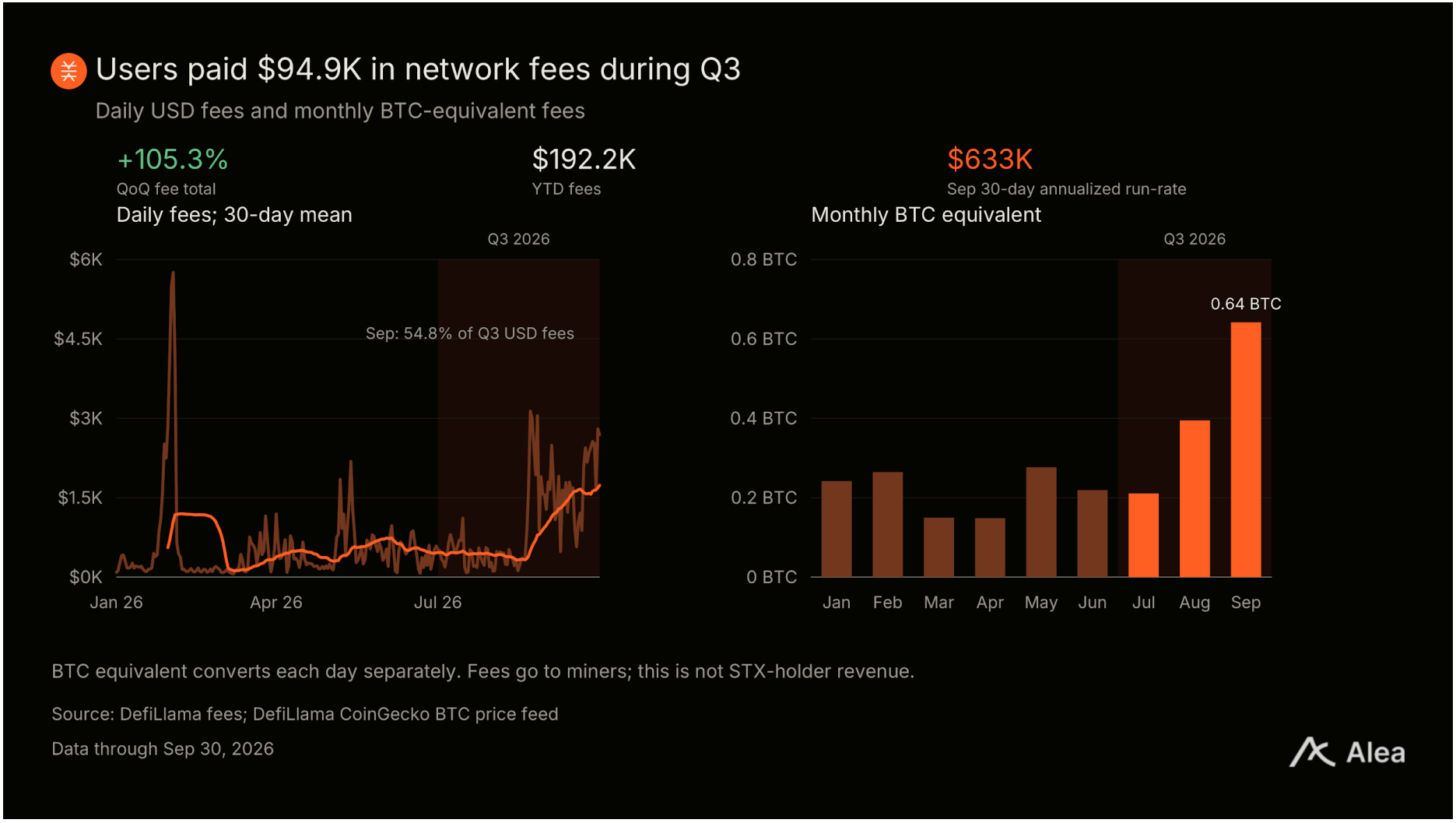


Stacks reported that block production and sBTC operations continued through the PoX-5 upgrade. The activity series counts contract calls, STX transfers, and deployments, including failed transactions.

Fees and Miner Economics

Fee recovery from a low base

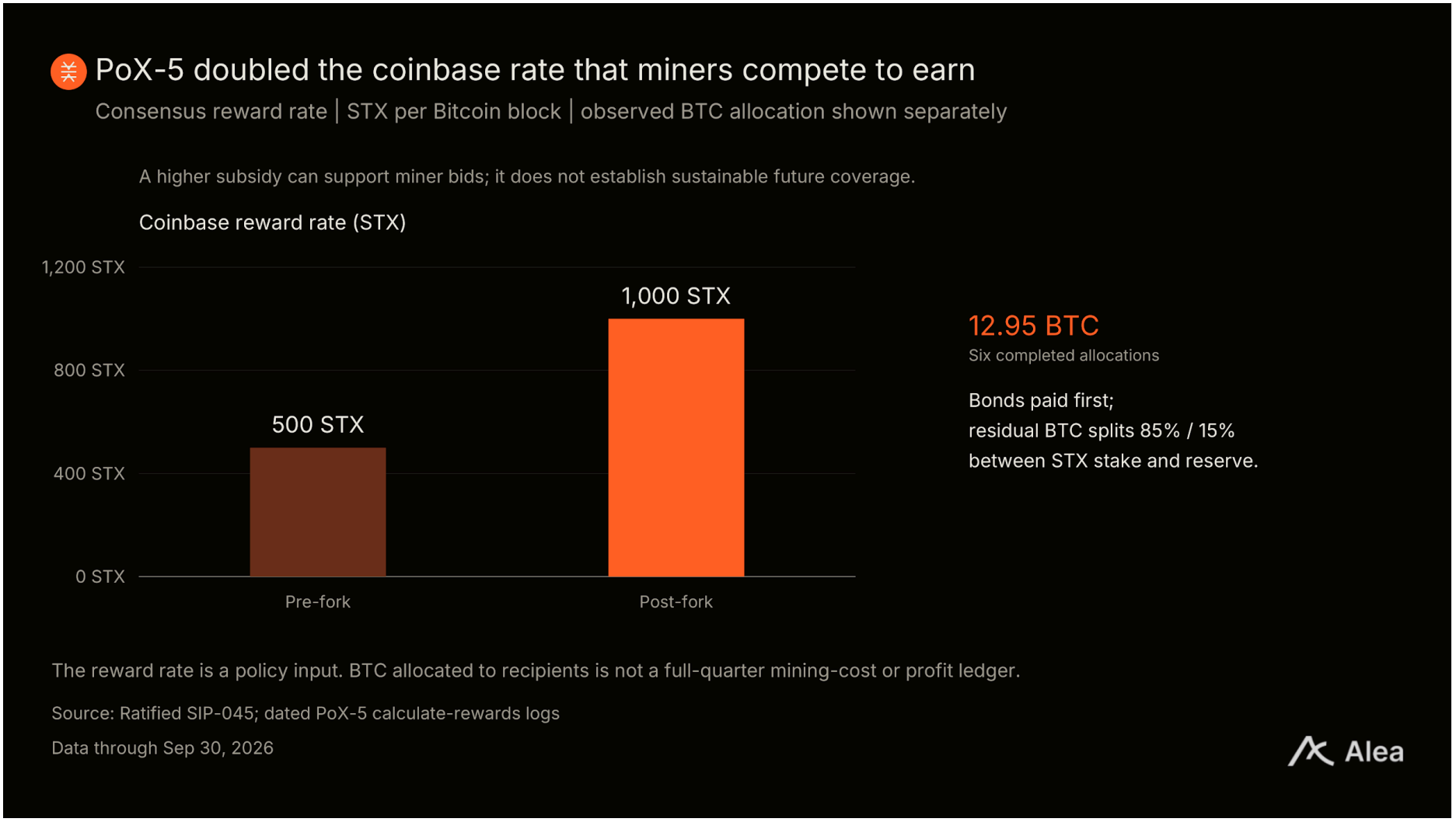
Users paid **$94.9K** in Q3 network fees, more than double Q2's **$46.2K** and up **167.5%** YoY. YTD fees totaled **$192.2K**. September's trailing-30-day average implied a **$633K** annualized run rate, above the **$376.5K** full-quarter annualization, reflecting a stronger quarter-end pace.



Fees are paid in STX, so dollar totals reflect token prices as well as transaction activity.

Issuance and reward funding

SIP-045 restored miner rewards from **500** to **1,000 STX** per Bitcoin block, doubling the STX subsidy during the bootstrap phase.



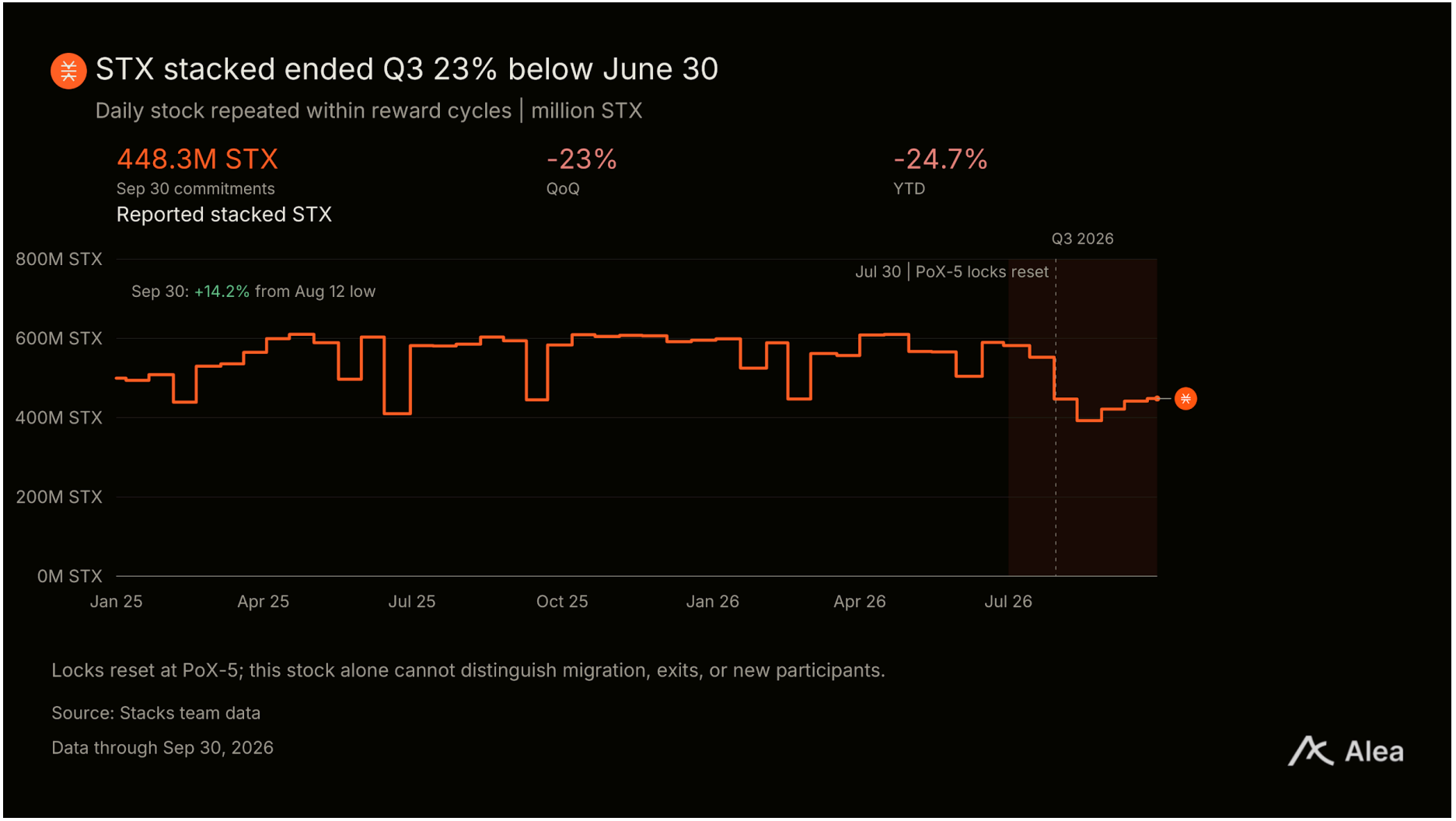
Six completed post-fork reward calculations allocated approximately **12.95 BTC** across recipients.

More STX issuance raises the mining reward available for BTC bids and dilutes existing holders.

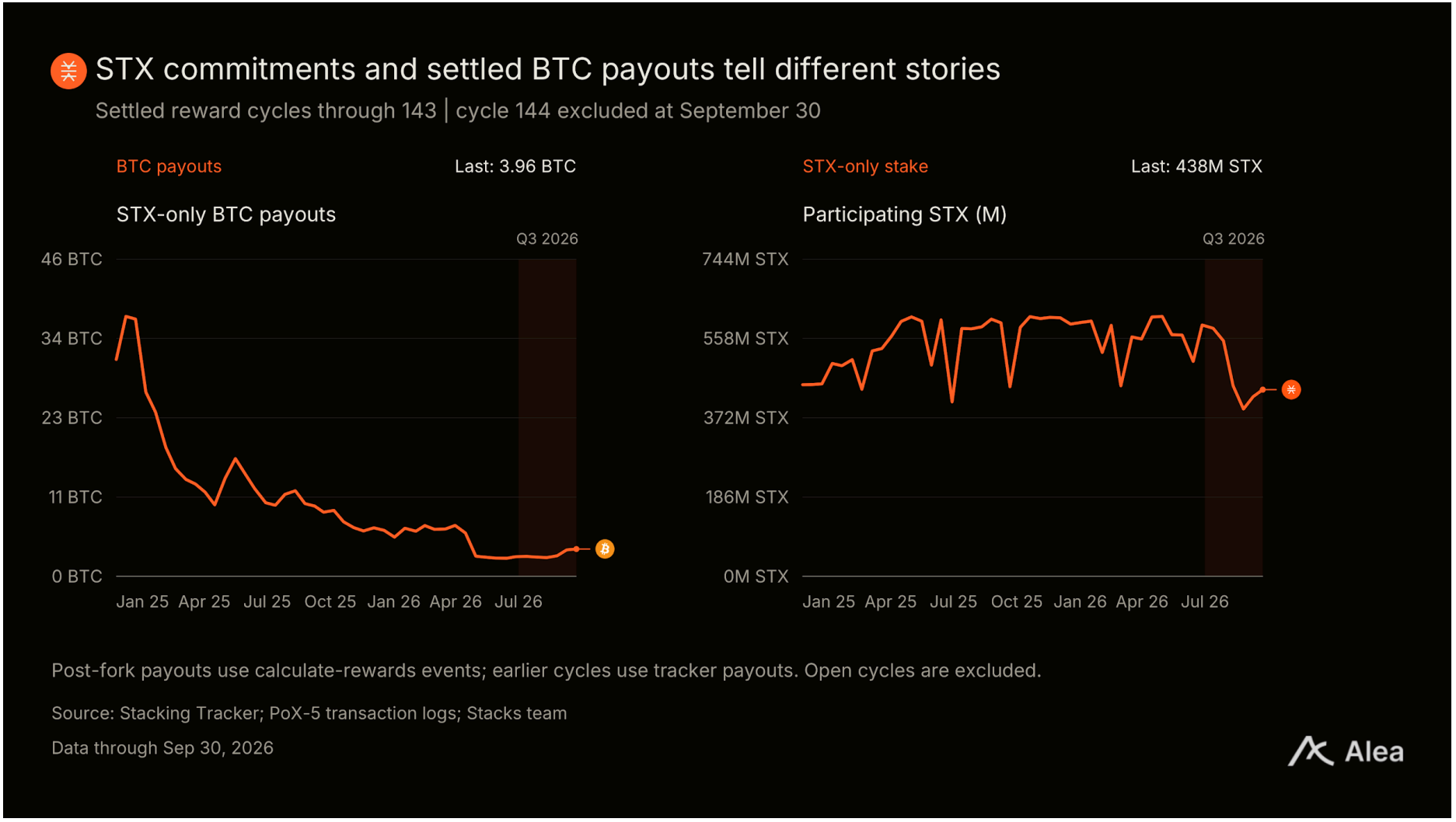
Stacking and Bitcoin Bonds

Stacking after the reset

Stacked STX recovered **14.2%** from its August 12 low to end September at **448.3M**, still **23%** below June 30's **582M**. PoX-5 unlocked existing positions and required restaking, so the quarter's decline includes contract migration.

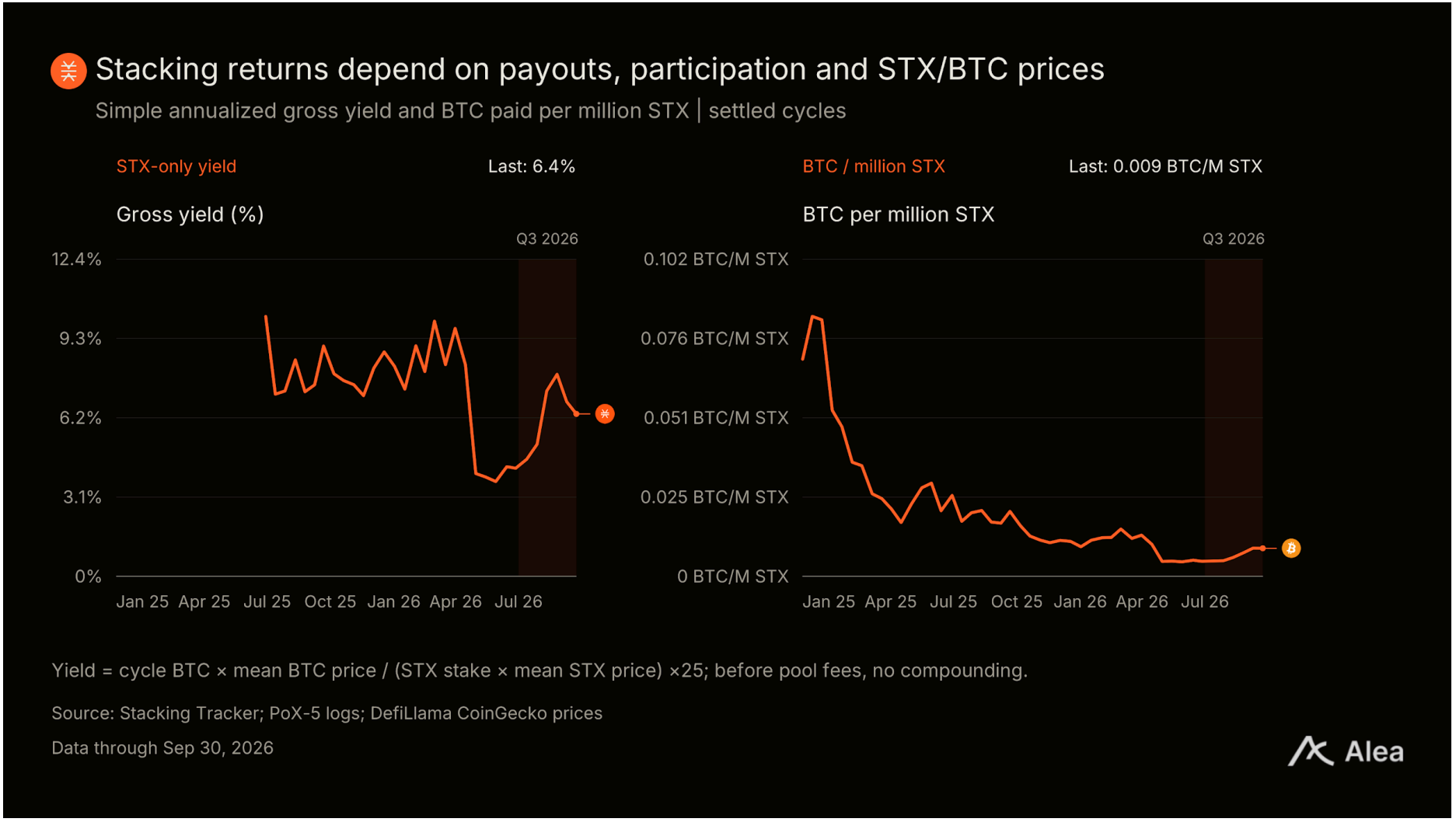


Cycle 143's realized-yield calculation uses **438M STX** of STX-only stake. Cycle 144 was still open at quarter-end and is excluded.



STX-only stacking returns

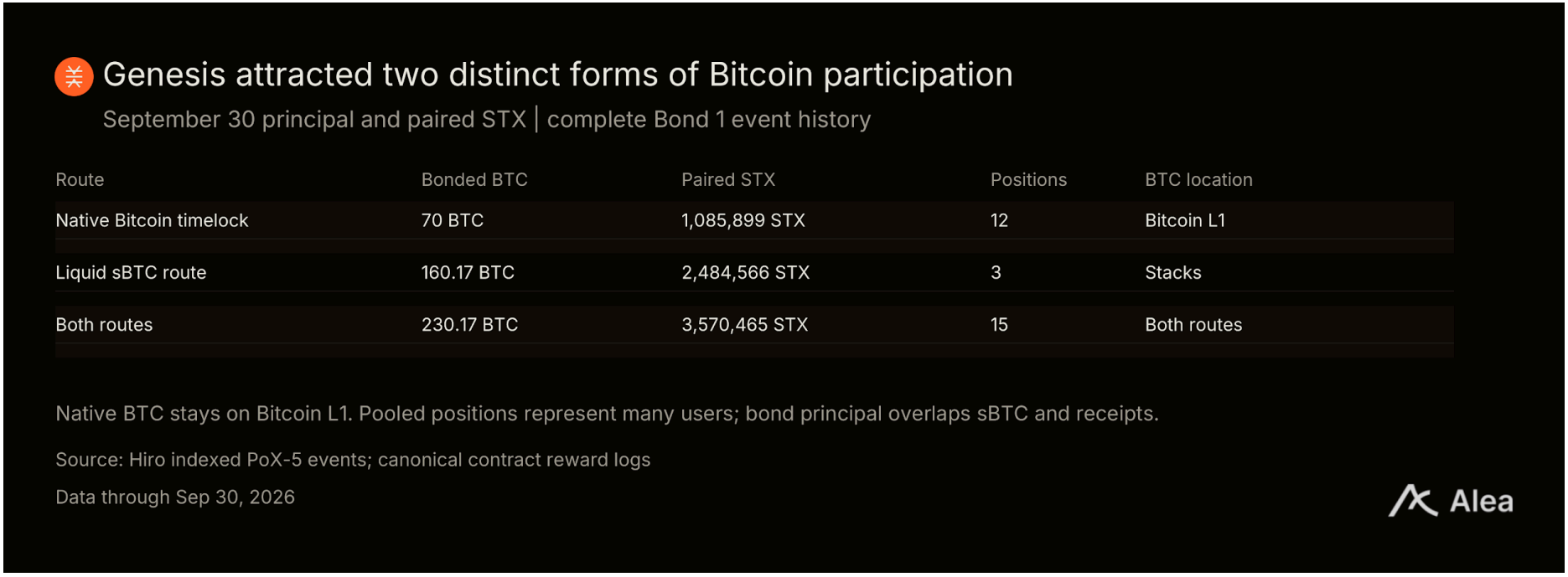
Cycle 143 allocated **3.96 BTC** to STX-only stackers, approximately **0.009 BTC** per million STX. Using the cycle's matched average price and multiplying the cycle return by 25 gives a **6.4%** simple annualized gross yield. The calculation excludes compounding and pool fees.



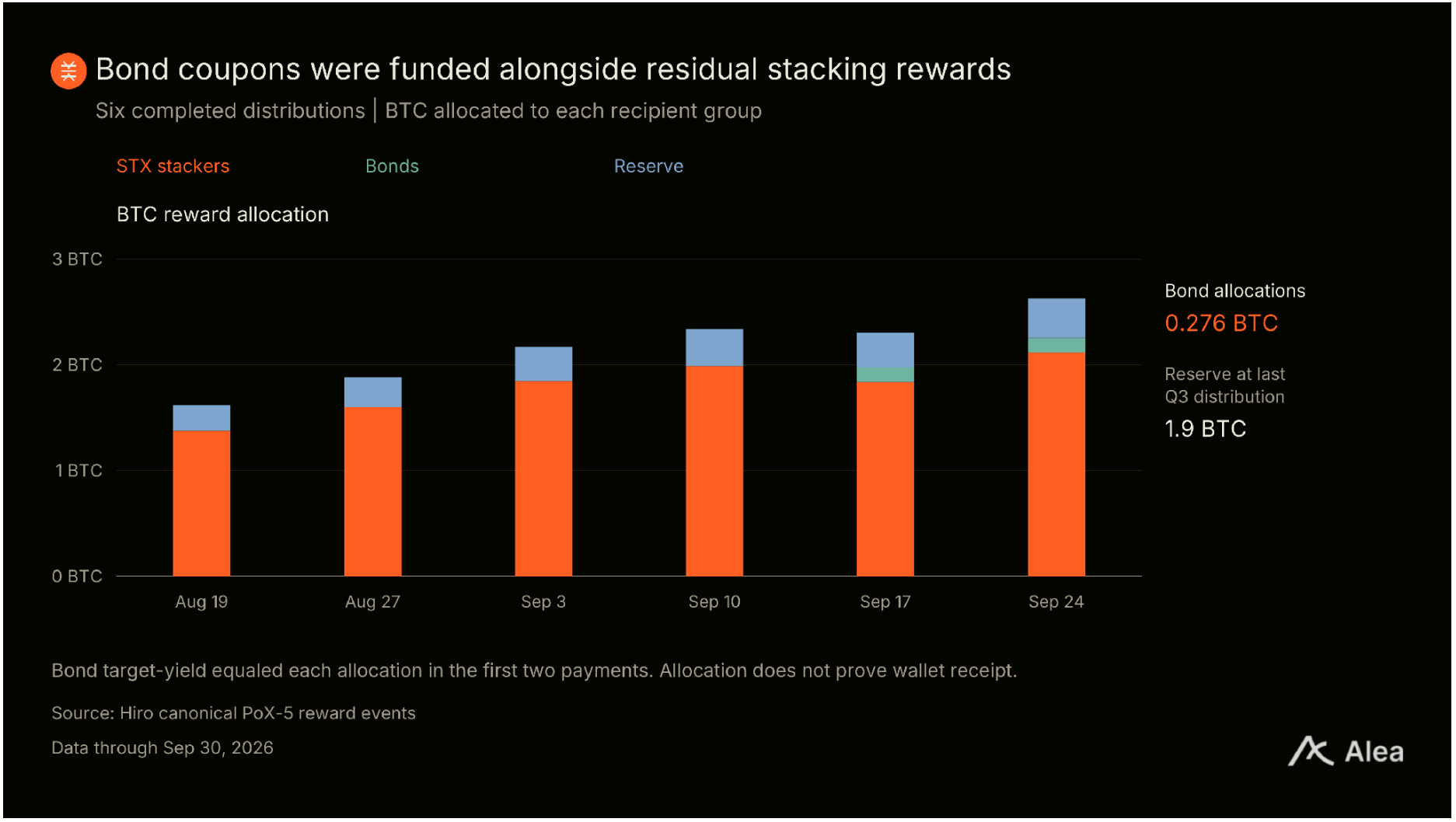
A smaller STX stake can raise rewards per token even when the BTC payout pool stays flat. STX prices also affect the dollar-value denominator.

Genesis demand and coupon coverage

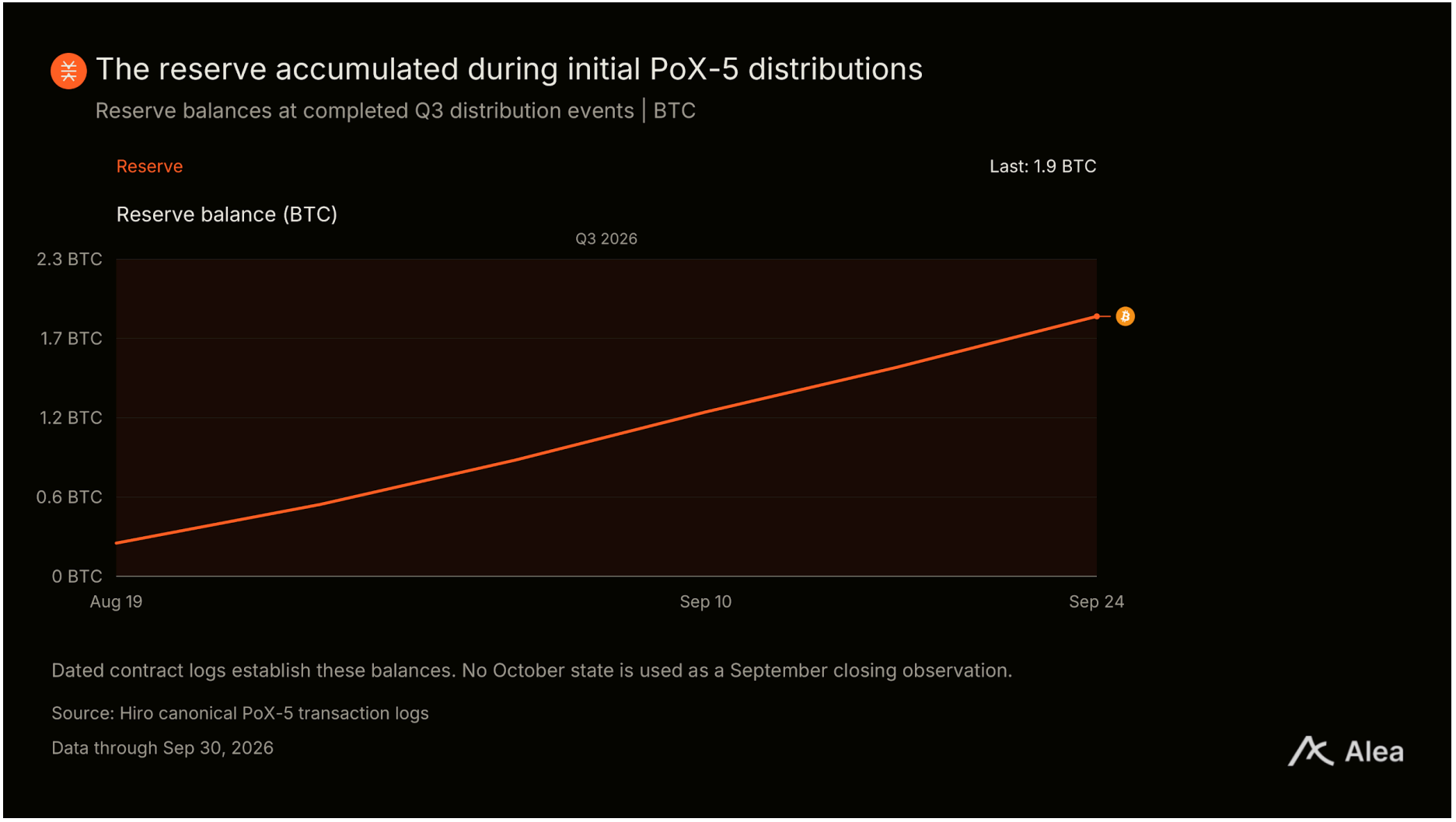
Genesis paired **230.17 BTC** with **3.57M STX**. Of the BTC principal, **69.6%** used the liquid route and **30.4%** remained in native Bitcoin timelocks. Paired STX stays locked for the bond term; it is not burned.



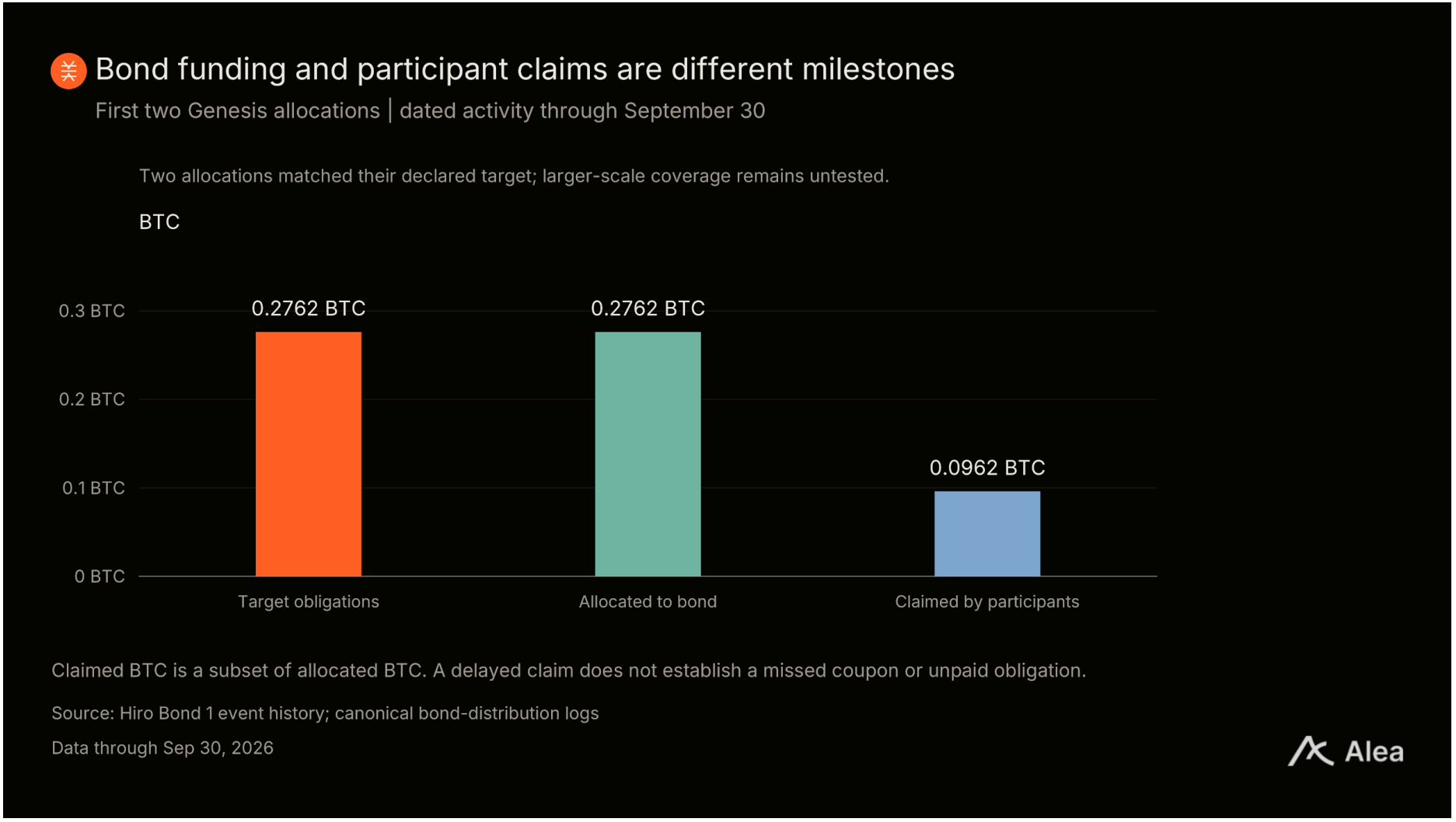
The first two bond reward calculations allocated **0.2762 BTC**, matching declared target obligations. Bonds receive priority, with residual BTC divided **85%** to STX-only stackers and **15%** to the reserve. Higher bond capacity therefore competes for the reward pool that previously supported STX-only participation.



The reserve held **1.9004 BTC** on September 24, the last completed Q3 distribution.



Indexed participant claims totaled **0.0962 BTC** by September 30. The tally tracks claim events rather than reward allocations.



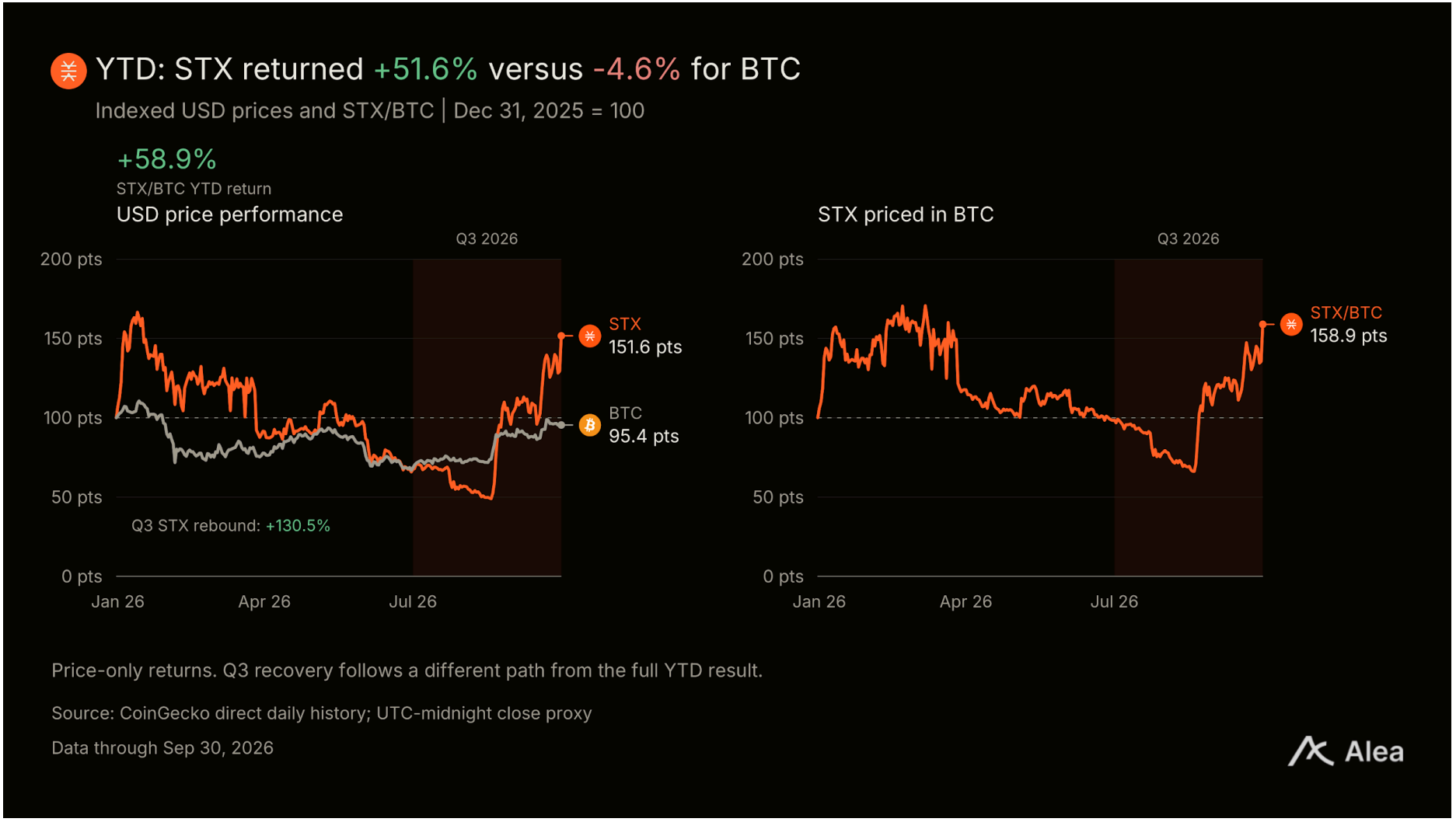
Market Data

STX outpaced BTC

STX returned **130.5%** in USD and **61.5%** against BTC during Q3.



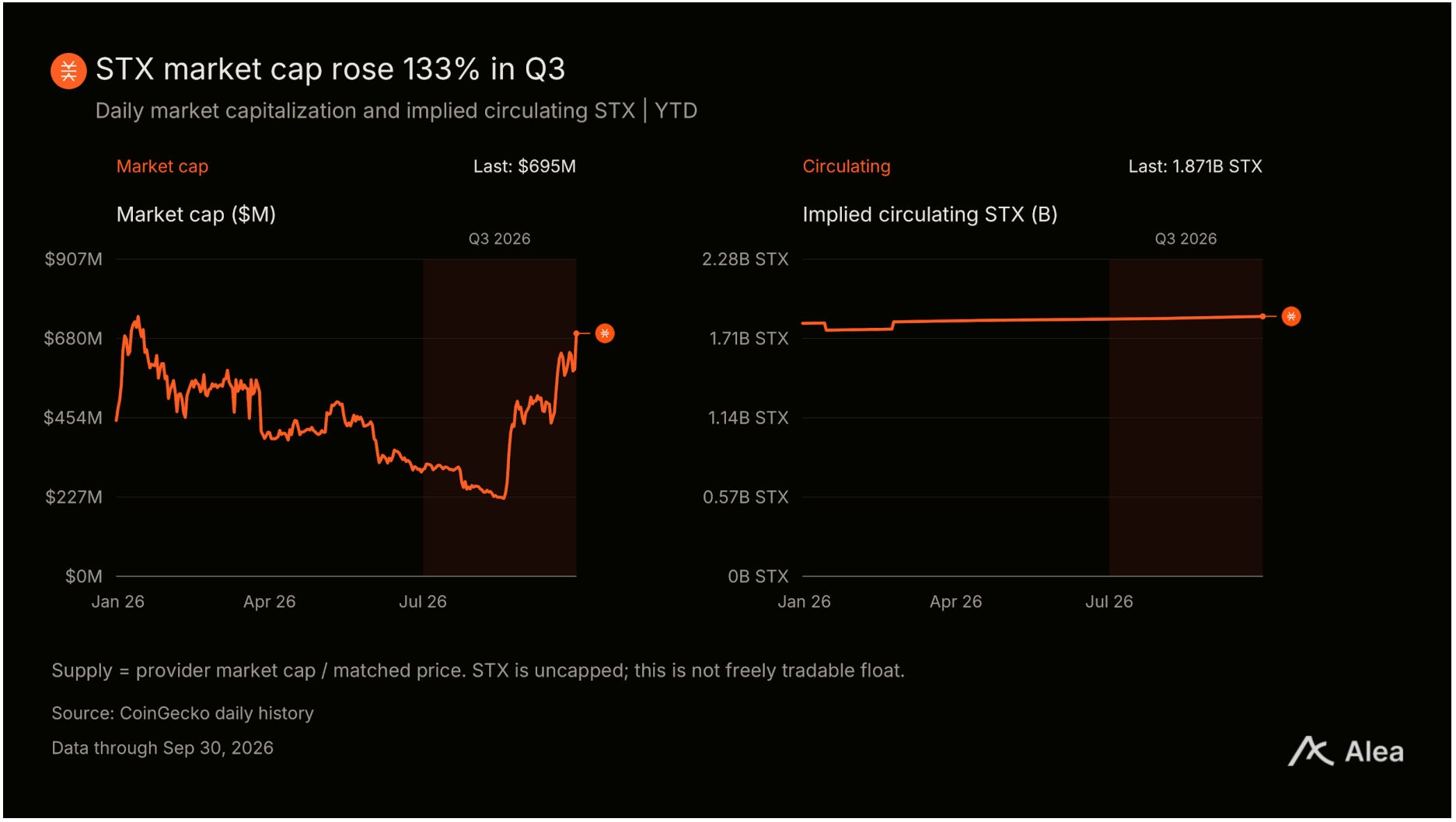
YTD in USD, STX gained **51.6%** while BTC fell **4.6%**.



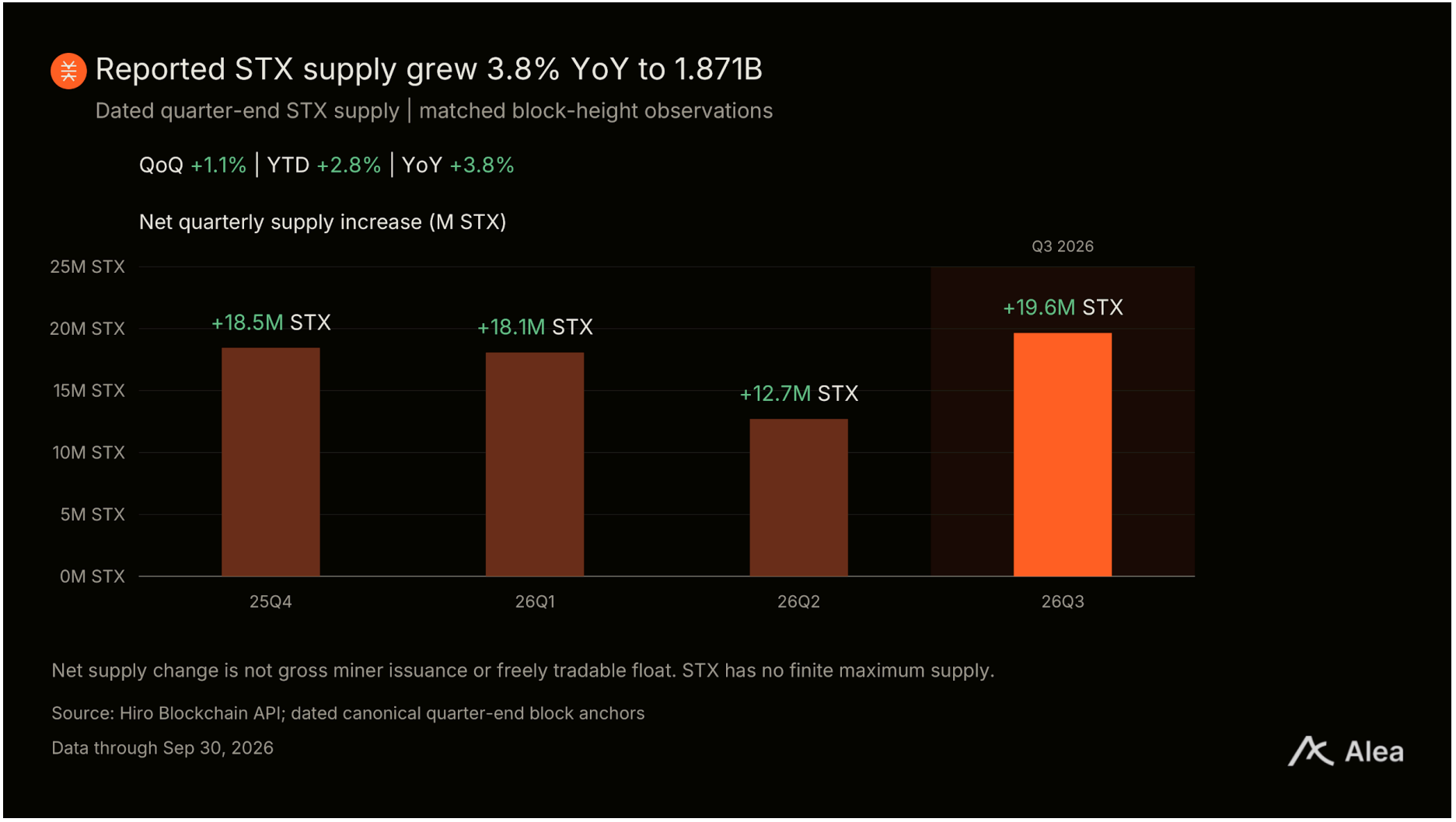
The Q3 rally recovered STX losses from earlier in the year. It coincided with Bitcoin Staking's launch and a BTC rally, while sBTC and stablecoin balances contracted.

Valuation and supply

STX market cap reached **$694.9M**, up approximately **133%** QoQ. Height-validated supply reached **1.871B STX**, increasing **19.65M** during Q3, or **1.1%**; supply was up **2.8%** YTD and **3.8%** YoY. Price appreciation explains most of the market-cap change.

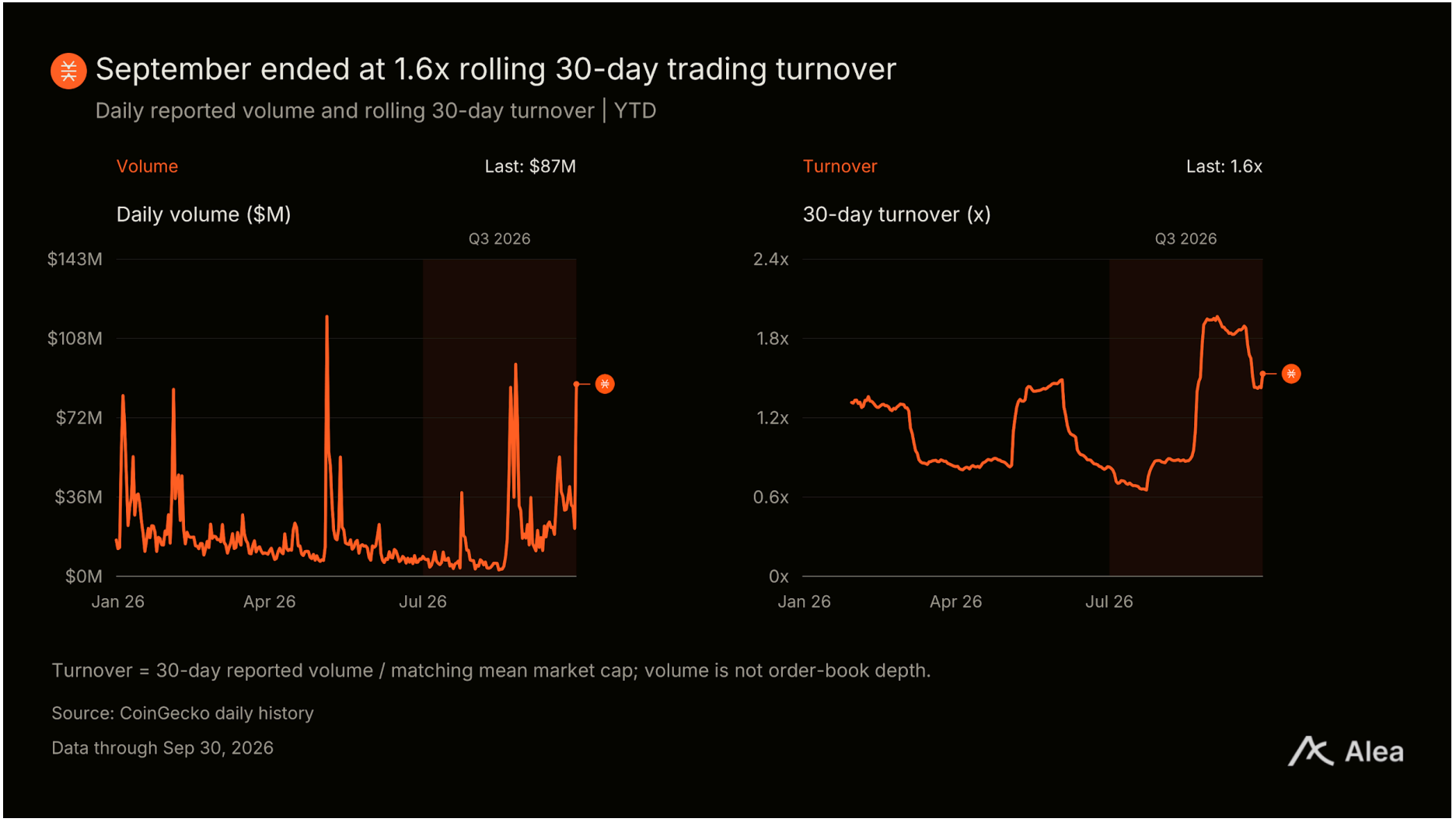


Genesis's **3.57M** paired STX was smaller than the quarter's **19.65M** net supply increase. Bonding locks tokens for a term; issuance increases total supply. STX supply is uncapped.

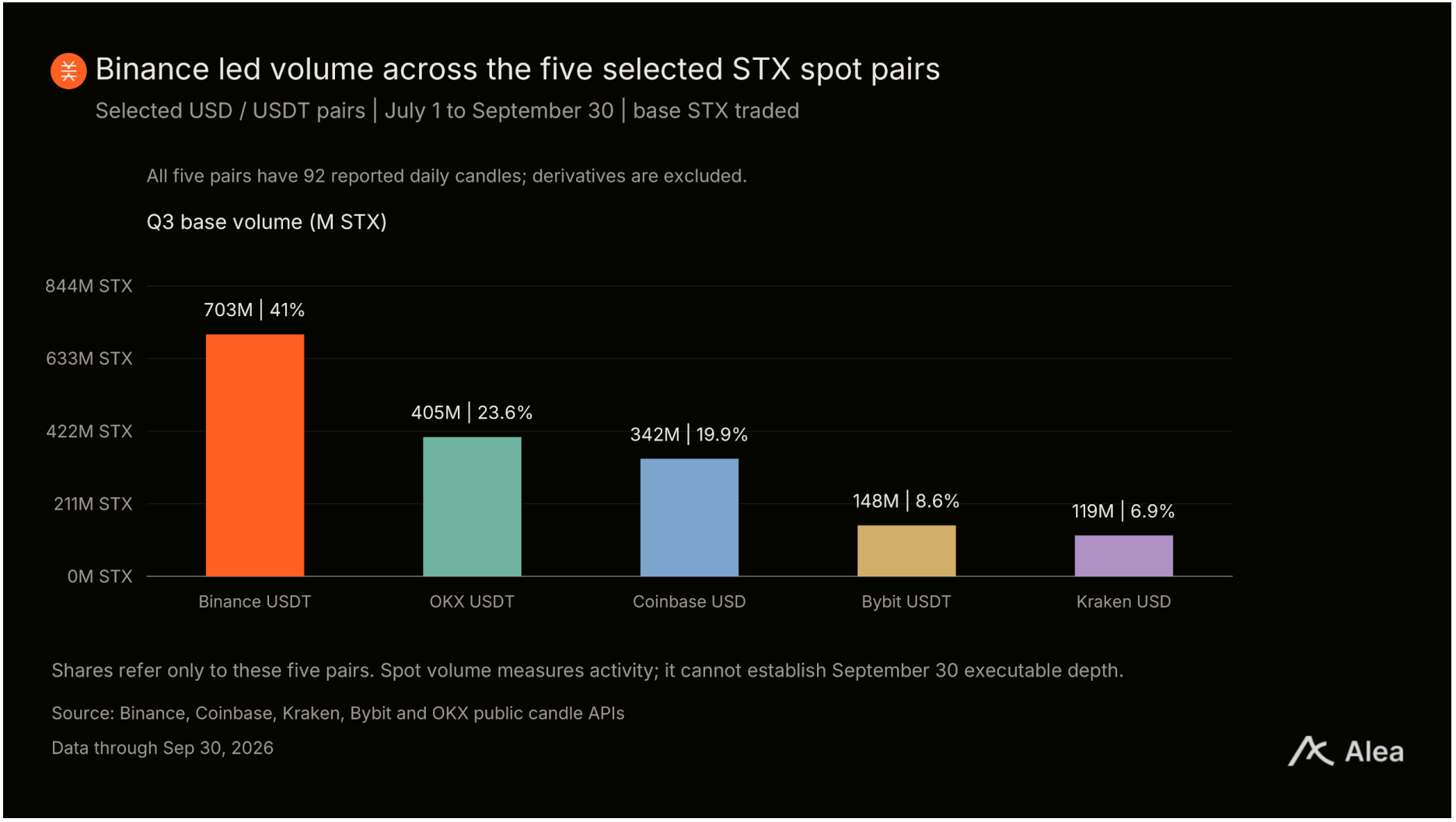


STX trading and turnover

STX trading volume totaled **$1.75B** during Q3, equivalent to **4.57x** average daily market cap. September's matching 30-day turnover was **1.56x**. Turnover measures trading relative to valuation, so a rising token price can offset growth in volume.

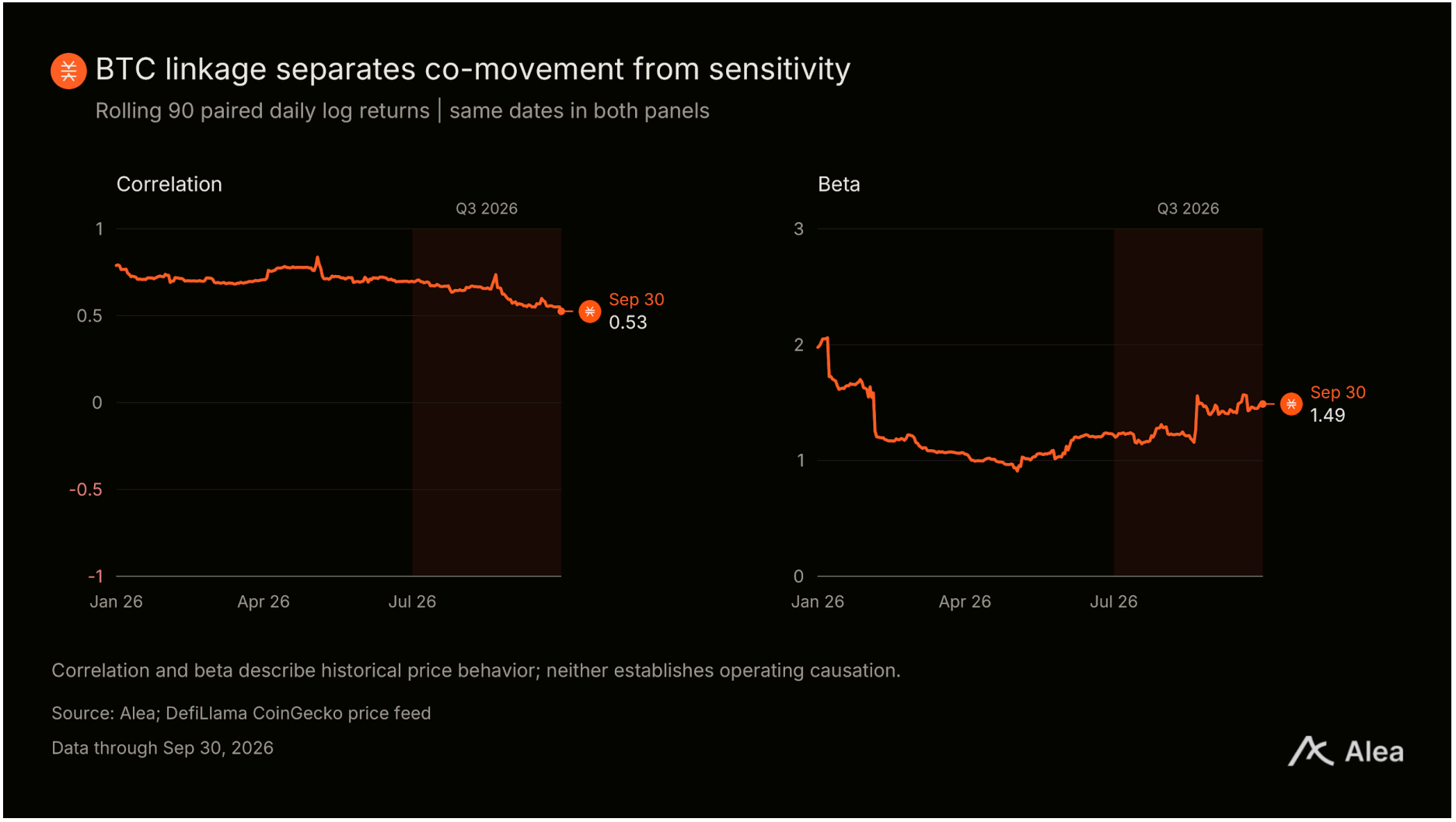


Across five selected spot pairs, Binance represented **41%** of aggregate base-token volume, followed by OKX at **23.6%** and Coinbase at **19.9%**. Binance applied a monitoring tag to STX on July 24, placing the token under closer listing review while trading and other services continued.



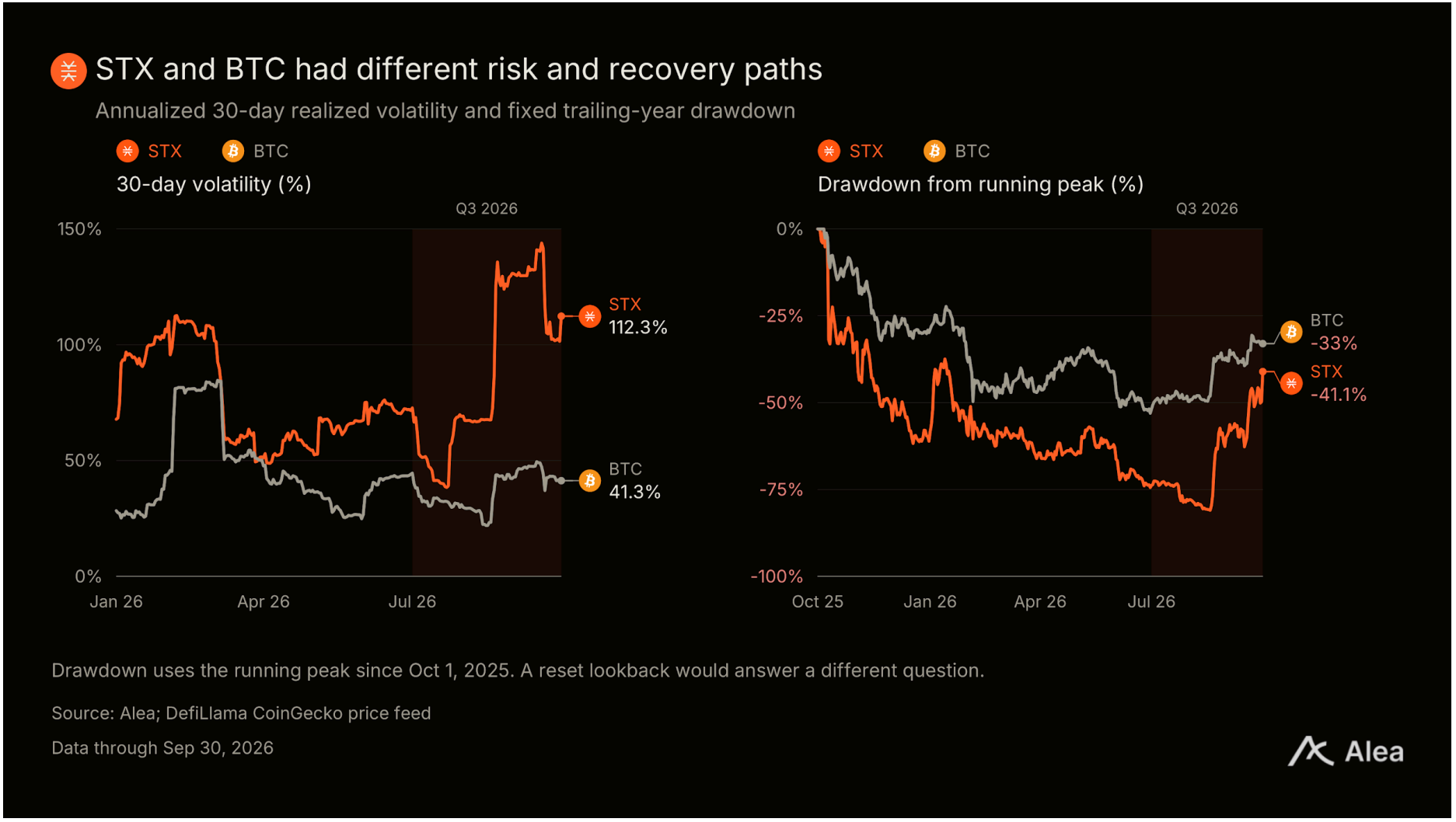
BTC sensitivity and risk

At quarter-end, matched rolling-90-day daily returns gave a **0.53** correlation and **1.49** beta to BTC. Beta captures STX's sensitivity to BTC over that window.



Trailing-30-day annualized volatility was **112.3%** for STX against **41.3%** for BTC.

From October 1, 2025 to September 30, 2026, STX remained **41.1%** below its observed peak versus **33%** for BTC. Q3's large gain coexisted with substantial prior losses.



Multiples

At September 30, STX's **$694.9M** market cap divided by a **$633K** trailing-30-day annualized fee run rate produced P/F of approximately **1,098x**. Network fees accrue to miners.



DeFi TVL represented **12.6%** of STX market cap, or **0.126x**. Market cap grew approximately **133%** QoQ against **10.3%** TVL growth, widening the gap between token valuation and application capital.

DeFi TVL equaled 12.6% of STX market value

DeFi TVL / STX market cap | YTD



A higher ratio can reflect more TVL or less market value; TVL does not belong to STX holders.

Source: DefiLlama; CoinGecko

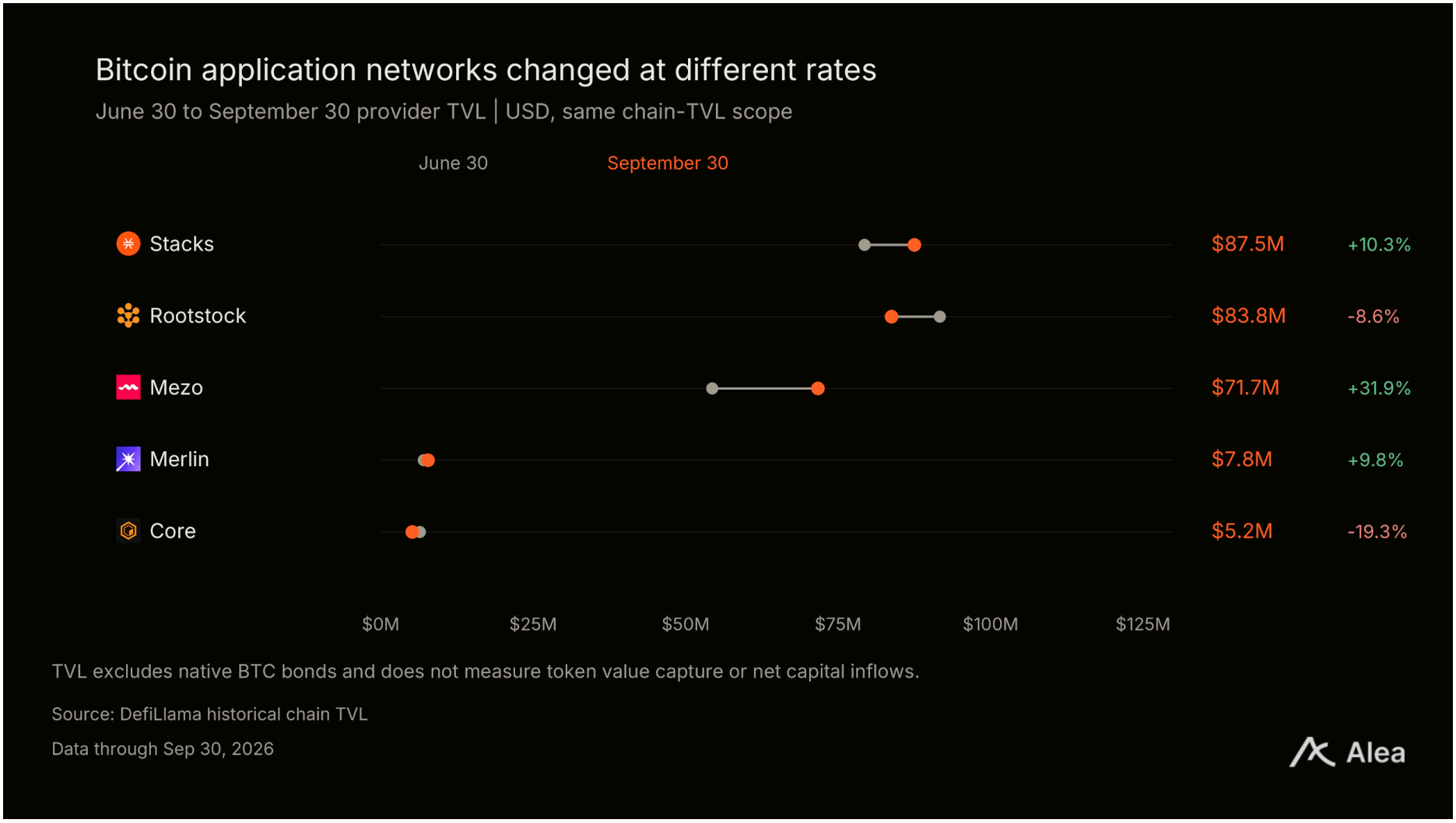
Data through Sep 30, 2026

Alea

Comps

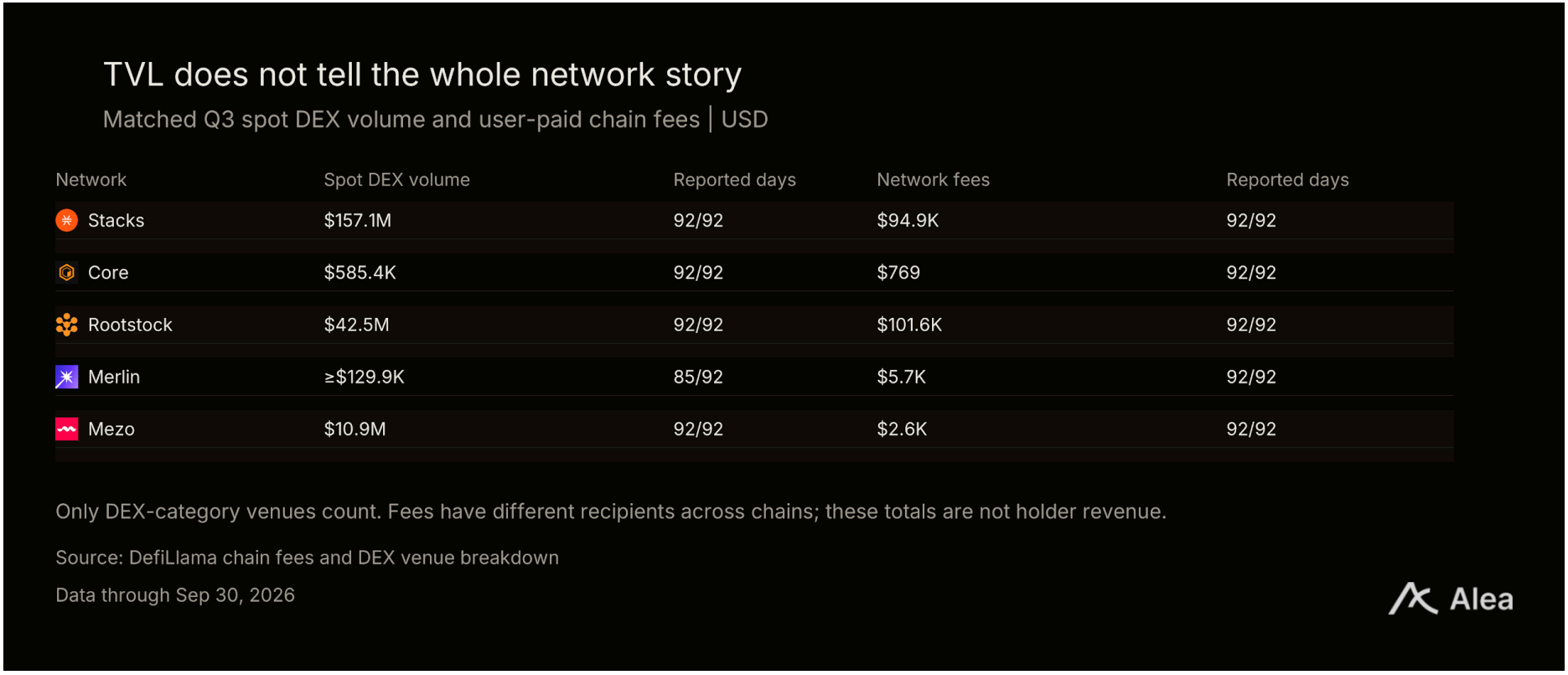
Network scale and activity

Within the five-network comparison, Stacks ended September with **$87.5M** TVL, ahead of Rootstock's **$83.8M** and Mezo's **$71.7M**. Quarter-average balances were much closer: **$81.69M** for Stacks and **$81.31M** for Rootstock. The endpoint ranking therefore overstates the size of the difference in capital present throughout Q3.



Mezo grew dollar TVL **31.9%** QoQ, versus **10.3%** for Stacks and **9.8%** for Merlin. Rootstock declined **8.6%**, and Core declined **19.3%**. The networks differ in asset and product mix.

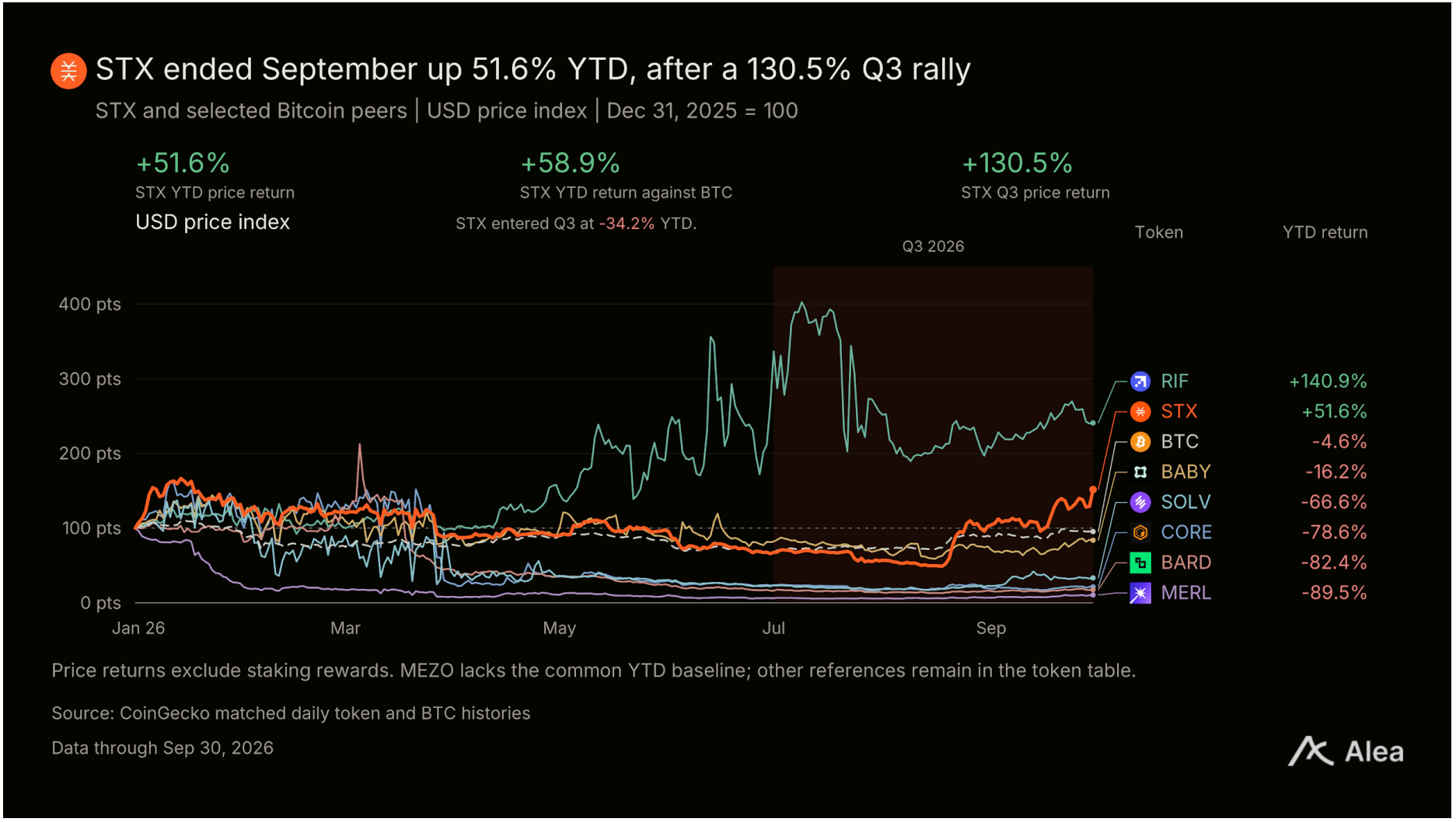
Stacks recorded **$157.1M** spot DEX volume, approximately **3.7x** Rootstock's **$42.5M**. Rootstock nevertheless recorded **$101.6K** of network fees, modestly above Stacks' **$94.9K**. Trading volume and network fee generation are therefore distinct competitive measures. Merlin's DEX history covers 85 of 92 days and is not treated as a fully matched quarter.



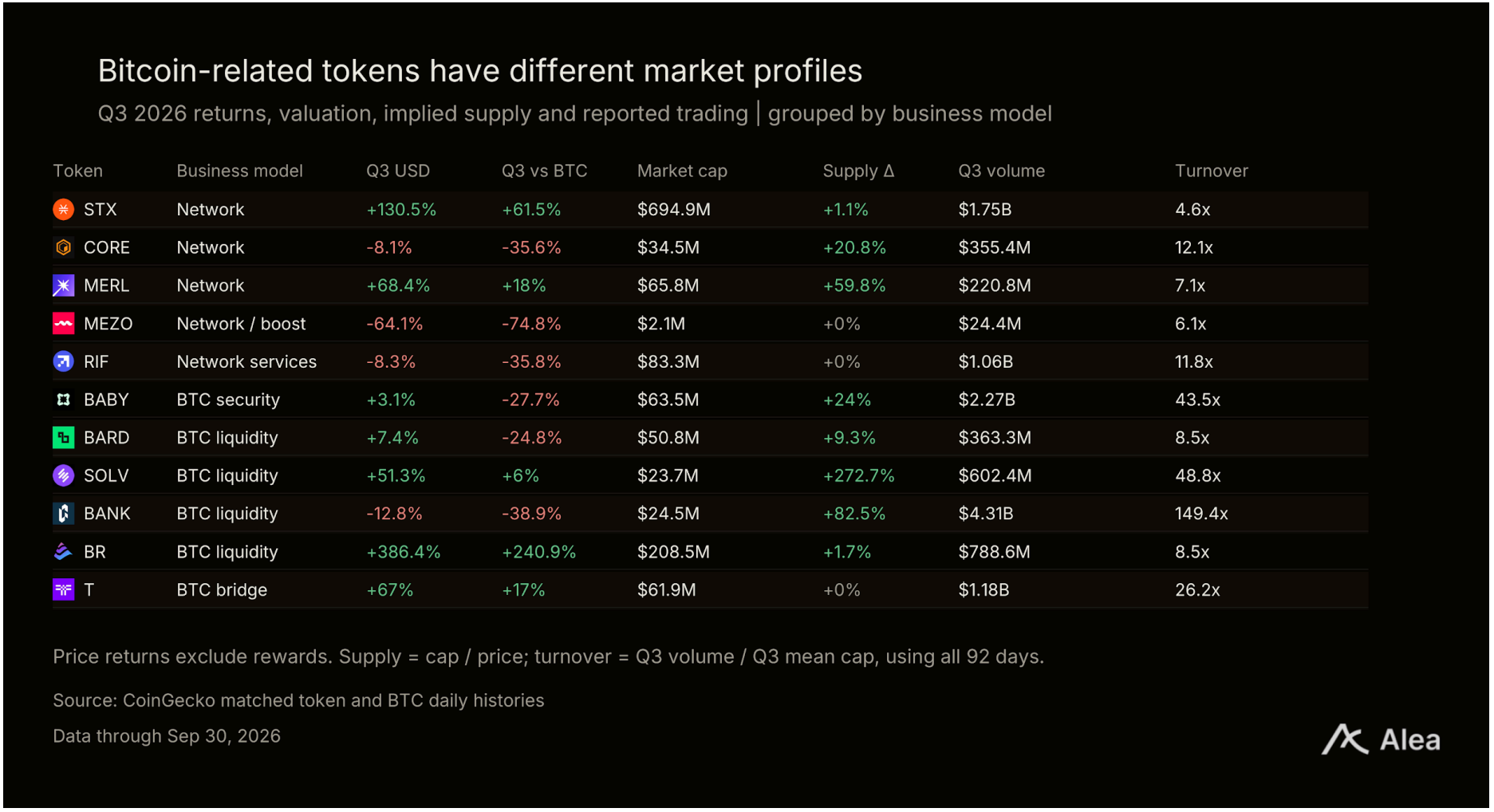
Stacks' volume lead was concentrated in Bitflow, which accounted for **97.8%** of quarterly trading.

Token returns and economics

The token panel separates networks from Bitcoin security and liquidity products. Bridging services form another group. STX gained **61.5%** against BTC and led the network-token subgroup on Q3 returns. Across the wider panel, BR returned **240.9%** against BTC, well ahead of STX. Positive USD performance did not always mean Bitcoin outperformance: BABY and BARD gained in dollars but fell **27.7%** and **24.8%**, respectively, against BTC.



In USD, STX entered Q3 down **34.2%** YTD and ended September up **51.6%**. RIF nevertheless gained **140.9%** YTD, so STX's lead among network tokens in Q3 did not extend across the year. The common baseline preserves earlier losses as well as the quarter's recovery. These price returns exclude staking rewards.



Implied circulating supply, calculated as market cap divided by price, changed substantially for SOLV and BANK. BANK's **149.4x** quarterly turnover measured reported trading against average market cap.

Token returns are not equivalent claims on protocol economics

Documented mechanisms | activated Q3 changes identified; no automatic cash-flow entitlement assumed

Token	Activity-to-token mechanism	Economic qualification
STX	Gas; paired bond capital; STX-only stacking	BTC paid to eligible stakers; network fees go to miners
CORE	Gas; validator staking; Dual Staking tiers	CORE rewards depend on active staking / validator
MERL	Governance; documented security staking	Prospective L3 fee utility is not current L1 revenue
MEZO	Lock to boost veBTC and direct gauges	Fees / incentives depend on locked BTC and voting
RIF	Services; stRIF governance participation	Rootstock gas uses rBTC; RIF has separate service utility
BABY	Gas; network staking; BTC co-staking	Staking rewards include token issuance
BARD	Governance; staking entry closed Sep 17	Existing staking rewards scheduled through Oct 1
SOLV	Governance; staking; redemption fee discounts	Emission rewards and discounts are distinct from profit
BANK	Governance / gauge utility described	Documentation excludes a promised revenue entitlement
BR	veBR governance and reward boosts described	Protocol activity does not guarantee holder cash flows
T	Governance; staked-T mint / redemption benefits	Fee waivers / rebates depend on participation

Documentation describes utility; enforceable returns depend on eligibility, actual deployment and reward funding.

Source: Official protocol token documentation; SIP-045; dated BARD closure notice

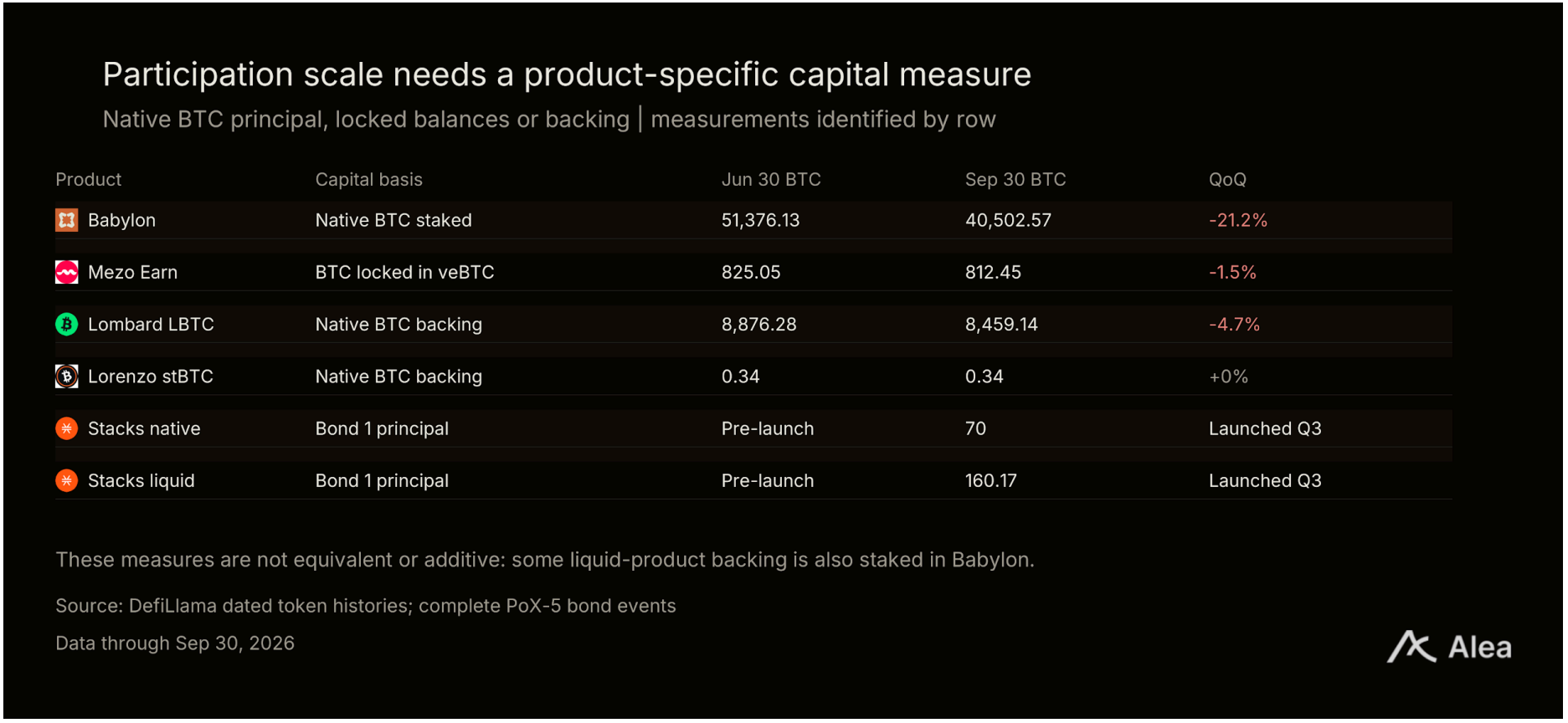
Data through Sep 30, 2026

Alea

STX's paired-bond requirement locks tokens for the bond term. Other tokens govern products or enable conditional reward access, giving holders different economic rights.

Bitcoin product capital

At September 30, Babylon had **40.5K BTC** staked. Lombard LBTC had **8.46K BTC** of native backing, and Mezo Earn had **812 BTC** locked. Stacks Genesis recorded **70 BTC** native and **160 BTC** liquid principal. Genesis had just 21 calendar days in its live reward period before quarter-end.

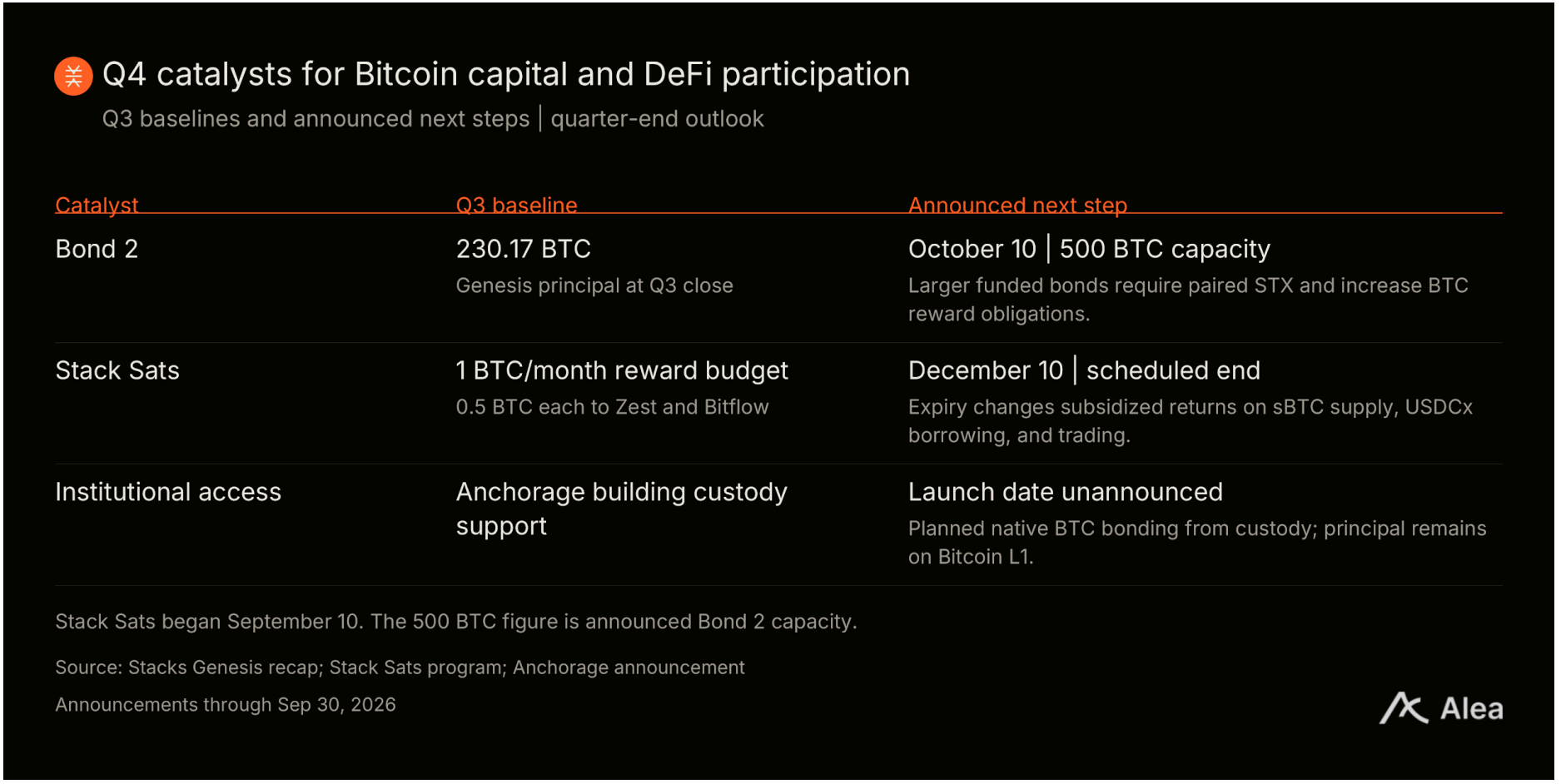


These balances measure different positions: native BTC staked, liquid-token backing, and timelocked BTC. Liquid-token backing can overlap with stake already counted in another product.

Conclusion

Q3 put Bitcoin Staking into operation and connected Genesis's liquid route to lending. Participants bonded **230.17 BTC** and paired **3.57M STX**; early reward allocations matched target obligations. With only 21 calendar days in the live Genesis reward period, the quarter established initial funding and product integration. Dollar TVL and fees recovered while sBTC supply and stablecoin balances fell, and the staking vault increased gross exposure through repeated borrowing and staking.

Bond 2 was scheduled for October 10 with **500 BTC** of capacity. Stack Sats allocates **1 BTC** per month to Zest and Bitflow through December 10, tying Q4 activity to a defined incentive window.



References

Stacks Website, Docs

DefiLlama

CoinGecko

Token Terminal

Hiro API, Stacking Tracker

Stacks PoX-5 activation, SIP-045, Genesis recap, Anchorage announcement, Stack Sats

Stacks Endowment Q3 community update

Zest Protocol September recap, Vault documentation, Liquidation rules

Binance Monitoring tag

Stacks team data: daily sBTC mint/burn flows, stablecoin supply, and stacked STX.

Disclosures

Alea Research is engaged in a commercial relationship with Stacks and receives compensation for research services. This report has been commissioned as part of this engagement.

Members of the Alea Research team, including analysts directly involved in this analysis, may hold positions in the tokens discussed.

This Content is Provided Solely for Informational and Educational Purposes. It does Not Constitute Financial, Investment, or Legal Advice.